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Systems and methods for controlling air properties in structures and inhibiting moisture accumulation and mold propagation in structures

Abstract

A structure comprises at least one outer wall having an internal wall section and an outer wall section with an air flow passage therebetween. A circulation system circulates air through the flow passage to inhibit moisture accumulation and mold growth. A sensing system determines the presence of moisture in the flow passage and generates a signal in response thereto. A controller receives the signal from the sensing system and controls the circulation system to maintain a predetermined temperature and relative humidity in the flow passage.

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Background/Summary

RELATED APPLICATIONS (1) This application is a Continuation-In-Part of U.S. patent application Ser. No. 17/556,527 filed 10 Dec. 2021 (Dec. 20, 2021), which is a continuation of U.S. patent application Ser. No. 16/273,344 filed 12 Feb. 2019 (Feb. 12, 2019) now U.S. Pat. No. 11,248,814 issued 15 Feb. 2022 (Feb. 15, 2022).

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

(1) Embodiments of the present disclosure relate to systems, apparatuses, and methods for reducing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior walls or exterior and interior walls of habitable structures.

(2) In particular, embodiments of the present disclosure relate to systems, apparatuses and methods

for reducing, removing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls of habitable structures, wherein the systems include air passageways or pathway supporting conditioned air through exterior walls or exterior and interior walls of habitable structures.

2. Description of the Related Art

(3) Previously, I presented a system and method for inhibiting moisture and mold in an outer wall structure, U.S. Pat. No. 7,247,090B2. I now present improved systems, apparatuses, and methods for reducing, inhibiting, or eliminating water and/or moisture penetration and/or accumulation in internal and external walls of buildings and for drying buildings that have been subjected to flooding or other water and/or moisture penetration and/or accumulation events.

SUMMARY OF THE DISCLOSURE

(4) Embodiments of present disclosure provide apparatuses and systems including exterior wall air passageways, optionally interior walls air passageways, and a heating, ventilation, and air conditioning unit (HVAC) that supplies conditioned air to the air passageways, valves, nozzles, and a size and a spacing of weep holes or air outlets associated with the walls and passageways or pathways to control the amount and direction of air flow into and through the passageways, wherein the conditioned air flow is designed to dry, to reduce moisture accumulation, and/or to inhibit mold growth in exterior walls or exterior and interior walls and/or to maintain a substantially uniform temperature in the structure throughout the day.

(5) Embodiments of present disclosure provide methods of drying, inhibiting moisture accumulation, or controlling moisture in exterior and/or interior walls of a structure and/or controlling or maintaining temperature, air flow and air quality within the structure, wherein the methods include providing exterior walls or exterior and interior walls having air flow passageways therein and supplying conditioned air into the flow passageway by the HVAC unit controlled by valves and/or nozzles to dry or reduce or inhibit moisture accumulation in the walls, to maintain a certain moisture content in the walls, and/or to maintain a substantially uniform temperature in interior spaces in the structure as well as maintaining air flow and air quality in the interior spaces. The term air quality here means maintaining, in all interior spaces, air flow, moisture content, temperature, oxygen content, carbon dioxide content, carbon monoxide contents at desired levels. It should be recognized that the air passageways or pathways may include valves and nozzles for controlling air flow into the passageways or pathways and the sizes, number, and spacing of weep holes or air outlets associated with the passageways and pathways control the flow of air out of the passageways or pathways. It should also be recognized that the valves may be “active” or “passive”, wherein the term passive means that the valves are one-way valves allowing flow into the passageways and pathways, but not out of the passageways or pathways, and wherein the term active means that the valves actively control the flow of air into the passageways or pathways. Again, the size, number and spacing of the passageway or pathway outlets along with valves ultimately control the air flow rate through the passageways or pathways.

Description

BRIEF DESCRIPTION OF THE DRAWINGS OF THE DISCLOSURE

(1) The disclosure may be better understood with reference to the following detailed description together with the appended illustrative drawings in which like elements are numbered the same:

(2) FIG. 1 depicts a prior art wall or building structure.

(3) FIG. 2A depicts a first embodiment of a wall or building structure of this disclosure.

(4) FIG. 2B depicts the first embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

(5) FIG. 3A depicts a second embodiment of a wall or building structure of this disclosure.

(6) FIG. 3B depicts the second embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

(7) FIGS. 4A&B depict expanded views of wall embodiments of this disclosure.

(8) FIG. 5A depicts a third embodiment of a wall or building structure of this disclosure.

(9) FIG. 5B depicts the third embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

(10) FIG. 5C depicts a fourth embodiment of a wall or building structure of this disclosure.

(11) FIG. 5D depicts the fourth embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

(12) FIG. 6A depicts a first embodiment of a multistory building structure of this disclosure.

(13) FIG. 6B depicts a second embodiment of a multistory building structure of this disclosure.

(14) FIG. 6C depicts a third embodiment of a multistory building structure of this disclosure.

(15) FIGS. 7A&B depict an embodiment of a peer and beam structure of this disclosure.

(16) FIGS. 8A-D depict fourth embodiments of wall air distribution systems of this disclosure.

(17) FIGS. 9A-C depict three other embodiments of wall air distribution systems of this disclosure.

(18) FIG. 10A depicts a fifth embodiment of a wall or building structure of this disclosure.

(19) FIG. 10B depicts the fifth embodiment of a wall or building structure of this disclosure with an attic exhaust assembly.

(20) FIGS. 11A&B depict expanded views of embodiments of outer wall assemblies of this disclosure, with sensors and a sprayer.

(21) FIGS. 12A&B depict expanded views of embodiments of outer wall assemblies of this disclosure, with sensors and a sprayer.

(22) FIG. 13 depicts a control system of this disclosure.

(23) FIGS. 14A-C depict three view of a window installation having an air flow pathway between the exterior and interior window panes, whether single exterior and interior pane windows or double exterior and interior pane window.

(24) FIG. 15A depicts a conventional HVAC system in a single family home.

(25) FIG. 15b depicts a conventional HVAC system including a heat exchanger in a single family home.

(26) FIG. 16A depicts an embodiment of a structural ventilation HVAC system of this disclosure.

(27) FIG. 16B depicts the structural ventilation HVAC system of FIG. 16A in a flooded state.

(28) FIG. 17 depicts an office building located in Houston, Texas that was subjected to analysis using the WUFI software program developed with assistance from Oak Ridge National Laboratory.

(29) FIG. 18A depicts a WUFI analysis of an attic assembly of a conventional HVAC system.

(30) FIG. 18B depicts a WUFI analysis of an attic assembly of an embodiment of a structural ventilation HVAC system of this disclosure.

(31) FIG. 19A depicts attic assembly temperatures and relative humidities at the top surface of the ceiling insulation of a conventional HVAC system.

(32) FIG. 19B depicts attic assembly temperatures and relative humidities at the top surface of the ceiling insulation of a structural ventilation HVAC system of this disclosure.

(33) FIG. 20A depicts a WUFI analysis of a wall assembly WUFI analysis of a conventional HVAC system.

(34) FIG. 20B depicts a WUFI analysis of a wall assembly WUFI analysis of a structural ventilation HVAC system of this disclosure.

(35) FIG. 21A depicts wall assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.

(36) FIG. 21B depicts wall assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.

(37) FIG. 22A depicts a WUFI analysis of a window assembly WUFI analysis of a conventional HVAC system.

- (38) FIG. 22B depicts a WUFI analysis of a window assembly WUFI analysis of a structural ventilation HVAC system of this disclosure.
- (39) FIG. 23A depicts window assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a conventional HVAC system.
- (40) FIG. 23B depicts window assembly temperatures and relative humidities at the exterior surface of the OSB sheathing of a structural ventilation HVAC system of this disclosure.
- (41) FIG. 24 depicts a school building located in Houston, Texas that was subjected to analysis using the WUFI software program developed with assistance from Oak Ridge National Laboratory.
- (42) FIG. 25A depicts the R.sub.HT of the attic assembly of the office building of FIG. 17.
- (43) FIG. 25B depicts the R.sub.HT of the attic assembly of the school building of FIG. 24.
- (44) FIG. 25C depicts the R.sub.HT of the wall assembly of the office building of FIG. 17.
- (45) FIG. 25D depicts the R.sub.HT of the wall assembly of the school building of FIG. 24.
- (46) FIG. 26 depicts the R.sub.HT of the window assembly of the office building of FIG. 17 and school building of FIG. 24.
- (47) FIG. 27 depicts the heating and cooling requirements of the building in various cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.
- (48) FIG. 28 depicts the annual heating energy usage of the office building at several cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.
- (49) FIG. 29 depicts the annual heating energy usage of the school building at several cities comparing conventional HVAC systems and the structural ventilation HVAC system of this disclosure.
- (50) FIGS. 30A-C depict moisture content readings at the walls of a one-story, one-year old, daycare.
- (51) FIGS. 31A-C depicts moisture content readings at the third floor walls of a three-story, three-year old, house.
- (52) FIGS. 32A-D depict moisture content readings at the third floor walls and ceiling of a three-story, four-year old, house.

DEFINITIONS USED IN THE DISCLOSURE

- (53) The term “at least one” means one or more or one or a plurality, additionally, these three terms may be used interchangeably within this application. For example, at least one device means one or more devices or one device and a plurality of devices.
- (54) The term “one or a plurality” means one item or a plurality of items.
- (55) The term “about” means that a value of a given quantity is within $\pm 20\%$ of the stated value. In other embodiments, the value is within $\pm 15\%$ of the stated value. In other embodiments, the value is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value.
- (56) The term “substantially” means that a value of a given quantity is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 2\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value. In other embodiments, the value is within $\pm 0.1\%$ of the stated value.
- (57) The term “HVAC” means a heating, ventilation, and air conditioning unit, which generally includes a furnace, cooling coils, and blower along with the electronic and electrical components that control the unit.

DETAILED DESCRIPTION OF THE DISCLOSURE

- (58) The inventor has found that new systems, apparatuses, and methods may be constructed and implemented for reducing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls of habitable structures by providing

conditioned air pathways in the walls of the structures that are externally vented to force the removal of water and/or moisture from the walls and reduce, remove, inhibit or eliminate water and/or moisture and/or their accumulation and/or mold growth in exterior and interior walls.

(59) The inventor has also found the new systems, apparatuses, and methods may include controlled structural ventilation of conditioned exhaust air from the attic, wherein the conditioned exhaust air comprises air that flow from the HVAC system into the interior of the structure and then is exhausted into the attic. A portion of the attic conditioned exhaust air may then distributed into the walls and between the window panes for energy saving and moisture remediation. In fact, the inventor surprisingly found that for existing structures using traditional HVAC units installing a controlled attic conditioned exhaust air vent. Such an attic control vent may afford up to about half the total energy saved that would result if a completed system were installed without the attic control vent designed to exhaust a portion of the conditioned exhaust air flowing into the attic from the interior of the existing structure into the surroundings. In addition to the energy savings, the cost of installation of the attic control vent is a fraction of the cost of an entire system. Thus, the installation of the attic control vent will unexpectedly result in energy savings for existing structures at a significant reduced installation cost compared to an entire system of this invention. Of course, it should be recognized that such an installation would not affect moisture reduction in exterior and/or interior walls and multi-planed windows.

(60) Embodiments of present disclosure broadly related to apparatuses and systems including: exterior wall air passageways; optionally interior walls air passageways; a heating, ventilation, air conditioning unit (HVAC) to supply conditioned air, and valves to control an amount and/or direction of air flowing into and through the passageways, wherein the conditioned air flow may dry the walls, reduce moisture content in the walls, inhibit or control mold growth in the walls, and/or maintain a substantially uniform temperature in the structure throughout the day and night.

(61) In certain embodiments, each exterior wall includes an internal wall section and an external wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, wherein the control valves or the unidirectional valves insure either a desired amount of air flow into the passageways or the air flow proceeds from the inlets to the outlets.

(62) In other embodiments, each exterior wall includes an internal wall section and an external wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, and each interior wall includes a first wall section and a second wall section with the passageway disposed therebetween, wherein each passageway includes an inlet having a control valve or a unidirectional valve and an outlet, wherein the control valves or the unidirectional valves insure either a desired amount of air flow into the passageways or the air flow proceeds from the inlets to the outlets.

(63) In other embodiments, the apparatuses and systems also include at least one sensor configured to sense moisture and generate a signal corresponding to the sensed moisture and a control unit configured to receive the signal from the at least one sensor and to control the HVAC and the valves to provide a conditioned air having properties to dry or reduce the sensed moisture or to maintain a certain moisture content in the walls due to the air flow into, through, and out of the passageways.

(64) In other embodiments, the apparatuses and systems also include at least one sensor disposed in one, some, or all of the passageways and configured to sense moisture in the passageways and to generate a signal corresponding to the sensed moisture and a control unit configured to receive the signal from the at least one moisture sensor and to control the HVAC and optionally the valves to provide conditioned air and control an amount of the conditioned air into the air flow passageways, wherein the conditioned air has properties to dry the walls or to reduce the sensed moisture in the walls or to maintain a certain moisture content in the walls as the conditioned air flows into, through, and out of the passageways.

(65) In other embodiments, the apparatuses and systems also include at least one sensor disposed in one, some, or all of the passageways and configured to sense temperature in the passageways and to generate a signal corresponding to the sensed temperature, and a control unit configured to receive the signals from the temperature sensors and to control the HVAC and optionally the valves to variably supply controlled amounts of conditioned air into the air flow passageways to maintain a constant and/or uniform temperature in the structure and/or to dry or reduce the moisture or to maintain a certain moisture content in the walls.

(66) In other embodiments, the apparatuses and systems also include a supply unit for supplying mold control compositions or other compositions into the walls, wherein the supply unit is under the control of the control unit. In other embodiments, the apparatuses and systems also include fire suppressant units that may feed carbon dioxide or carbon dioxide foam or other fire retardant or suppression compositions.

(67) In other embodiments, the apparatuses and systems are designed to permit conditioned air to be flow through the exterior and interior walls so that a uniform temperature and moisture content in the wall interiors may be maintained. During the day and night, the apparatuses and systems may continuously adjust rate and amount of conditioned air flowing into each passageway so that the temperature and moisture content within the interior of each wall is maintained at substantially uniform values that may be preset set or set as desired by the user. As the sun rises, one side of a structure will begin to heat differently than the non-illuminated side, the apparatuses and systems adjust the distribution of air flow so that the temperature and/or moisture content in the interior the walls and therefore throughout the entire structure may be more uniformly controlled.

(68) Embodiments of present disclosure broadly relates to methods of drying water or moisture in exterior walls or exterior and interior walls of a structure, inhibiting moisture accumulation in the walls, or maintaining a substantially uniform temperature and/or moisture content in the wall, where the methods include providing exterior walls or exterior and interior walls having air flow passageways and supplying conditioned air into the flow passageway by an HVAC unit and valves to dry or reduce or inhibit moisture accumulation in the walls, to maintain a certain moisture content in the walls, and/or to maintain a substantially uniform temperature in the structure.

(69) In certain embodiments, the methods further include receiving output from at least one sensor and controlling properties of the conditioned air in response thereto.

(70) In other embodiments, the methods further include receiving output from at least one sensor, controlling properties of the conditioned air in response thereto and controlling an air flow rate into the passageways in response thereto.

(71) In other embodiments, the methods further include supplying a treating composition into the passageways, wherein the treating composition is a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a poison for rodents, other control compositions, or mixtures and combinations thereof.

(72) In other embodiments, the methods also include collecting temperature data, moisture data, air flow data, pressure data, air component data, and/or other data and adjusting the air flow rate and/or amount into each passageway so that the values of each measure property may be maintained at some desired level or some pre-set level. In other embodiments, the adjusting may be in conjunction with data indicating a fire or other emergency situation in which case the air flow and fire suppressants and/or retardants flow rate into the passageways may be used to impede progression of the fire or to produce fire breaks, wall as that do not support fire due to air flow and flame retardant flow rate into each passageway.

(73) Again, it should be recognized that the air passageways or pathways may include valves and nozzles for controlling air flow into the passageways or pathways and the sizes, number, and spacing of weep holes or air outlets associated with the passageways and pathways control the flow of air out of the passageways or pathways. It should also be recognized that the valves may be “active” or “passive”, wherein the term passive means that the valves are one way valves allowing

flow into the passageways and pathways, but not out of the passageways or pathways, and wherein the term active means that the valves actively control the flow of air into the passageways or pathways. Again, the size, number and spacing of the passageway or pathway outlets along with valves ultimately control the air flow rate through the passageways or pathways.

(74) Embodiments of this disclosure broadly relate to structures including a floor or slab, a ceiling, an insulated roof, and an attic interposed between the ceiling and the roof. The structures further include one or a plurality of exterior walls, each of the exterior walls includes: an exterior wall outer portion, an exterior wall inner portion including a wall framing system, an exterior wall air flow passageway interposed therebetween, and at least one exterior wall exhaust conduit having: a passageway inlet, and a passage way outlet. The structures further include a plurality of interior walls and a plurality of interior spaces defined by the exterior and interior walls. The structures further include an HVAC unit, disposed in the attic, including: an HVAC air inlet conduit having: an HVAC inlet, and an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit; an HVAC air outlet conduit having: an HVAC outlet, and an HVAC outlet control valve configured to control a supply of conditioned air to the interior spaces; and at least one HVAC exhaust conduit having: an HVAC exhaust inlet, and an HVAC exhaust outlet configured to support an air flow of exhaust conditioned air into the attic. The structures further include at least one passageway control valve associated with the passageway inlets and configured to support passageway air flow into the passageway(s) and out of the exhaust conduit(s) into the surroundings and to stop air flow from the surroundings into the passageways, whereby the passageway air flow reduces, removes, and/or eliminates moisture in the exterior walls.

(75) In certain embodiments, the at least one passageway control valve is a passive one-way or unidirectional valve, and the passageway air flow comprises attic exhaust conditioned air. In other embodiments, the at least one passageway control valve is part of the HVAC unit and is associated with a HVAC passageway outlet conduit connected to the HVAC outlet conduit and having a passageway outlet associated with the passageway inlets, the at least one passageway control valve is a passive one-way or unidirectional valve or an active control valve, and the passageway air flow comprises conditioned air from the HVAC unit. In other embodiments, each of the interior walls includes: two interior wall outer portions, an interior wall inner portion having: an interior wall air flow passageway having: a passageway inlet and a passageway outlet; at least one interior wall exhaust conduit. In other embodiments, the exterior wall inner portion includes: an insulating layer, a weather-resistant layer disposed on an outer surface of the insulating layer; and/or a sheathing disposed on an inner surface of the insulating layer. In other embodiments, the exterior wall inner portion further includes: a liquid barrier and/or a first sheathing between the liquid barrier and the wall framing system. In other embodiments, the at least one HVAC exhaust conduit further includes: a second HVAC exhaust outlet configured to support an air flow of exhaust conditioned air into the surrounding and having: an HVAC exhaust control valve configured to control amounts of exhaust conditioned air exhausted into the attic and into the surrounds.

(76) In other embodiments, the structure further comprising: an air flow control system including: a processing unit; and plurality of sensors; wherein one, some, or all of the control valves are active control valves, wherein the processing unit is in communication with the active control valves, and wherein the processing unit is configured to: receive input from the sensors, and control the HVAC unit and the active control valves thereby controlling air flow rates into the interior spaces and into the passageways. In other embodiments, the sensors are selected from the group consisting of temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof. In other embodiments, at least one of the sensors is disposed in: the air flow passageways or the air flow passageways and the attic and/or the interior spaces. In other embodiments, all of the control valves are active control valves, and during a 24 hour period of time, the processing unit is further configured: to adjust the air flow into each of the interior

spaces and into each passageway so that one or more interior space properties and/or one or more passageway properties are maintained at desired values. In other embodiments, the one or more properties include moisture concentrations in the interior spaces and/or passageways, temperature in the interior spaces and/or passageways, or mixtures and combinations thereof. In other embodiments, the one or more properties further include pressures in the interior spaces and/or passageways, air flow rates in the interior spaces and/or passageways, sensed chemical concentrations in the interior spaces and/or passageways, and/or microbial concentrations in the interior spaces and/or passageways.

(77) In other embodiments, the structures further comprises: a dispensing assembly associated with one, some, or all of the passageways and including: a reservoir containing a treating composition; a dispensing supply conduit having: a dispensing assembly control valve, a dispensing head at its distal end, wherein the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the passageways. In other embodiments, the treating composition is selected from the group consisting of a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, and mixtures or combinations thereof, and the gas is selected from the group consisting of nitrogen, nitrogen enriched air, argon, haloalkanes and mixtures thereof, the haloalkanes are selected from the group consisting of iodomethane (Halon 10001); bromomethane (Halon 1001), bromochloromethane, CH_2BrCl (Halon 1011), carbon tetrachloride (Halon 104), tribromofluoromethane (Halon 1103), dichlorofluoromethane (Halon 112), bromodifluoromethane (Halon 1201), dibromodifluoromethane (Halon 1202), bromochlorodifluoromethane, CF_2ClBr (Halon 1211), dichlorodifluoromethane (Halon 122), bromotrifluoromethane, CBrF_3 (Halon 1301), tetrafluoromethane (Halon 14), 1,2-dichlorotetrafluoroethane (Halon 242), dibromotetrafluoroethane, $\text{C}_2\text{Br}_2\text{F}_4$ (Halon 2402), hexafluoroethane (Halon 2600), and mixtures thereof.

(78) Embodiments of this disclosure multistory structures comprising: a plurality of stories, each of the stories including: a floor, a ceiling, and a dead spaces interposed therebetween. The multi-story structures further include: a plurality of exterior walls, each of the exterior walls includes: an exterior wall outer portion; an exterior wall inner portion; an exterior wall framing system; an exterior wall air flow passageway interposed between the exterior wall outer portion and the exterior wall inner portion; and at least one exterior wall exhaust conduit. The multi-story further include: a plurality of interior walls; a plurality of interior spaces defined by the exterior walls and interior walls; and an air flow control system including: a processing unit; plurality of sensors. The multi-story structures further include: an air handling system comprising: an HVAC unit; an HVAC inlet conduit including: an HVAC surroundings inlet conduit having: a surrounding inlet; and a surrounding inlet control valve configured to supply unconditioned air to the HVAC unit; a return conditioned air conduit having: a return conduit control valve configured to control an amount of exhausted conditioned air to the HVAC unit; a plurality of floor return conduits, each of the floor return conduits includes: an inlet disposed in the each of the floor dead spaces; a floor return conduit control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the floor dead spaces; and a main HVAC outlet conduit including: a plurality of floor outlet conduits, each of the floor outlet conduits includes: a floor outlet conduit control valve configured to control an amount of conditioned air supplied to each of the floors; a plurality of interior space outlet conduits, each of the interior space outlet conduits having: a plurality of interior space outlets, each of the interior space outlets; and an interior space outlet conduit control valve configured to control an amount of conditioned air supplied to each of the interior space outlets, wherein one, some, or all of the control valves are active control valves, wherein at one or more of the sensors is disposed in each of the interior spaces and one or more of the sensors is disposed in the passageways, wherein the processing unit is in communication with the active control valves, the HVAC unit, and the sensors, and wherein the processing unit is

configured to: receive input from the sensors, and control the HVAC unit and the active control valves based on the sensor inputs, thereby controlling unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings.

(79) In certain embodiments, each of the floor return conduits further includes: at least one return air control valve configured to control an amount of the return conditioned air to the return conduit.

(80) Embodiments of the disclosure broadly relates to methods for controlling air flow in a single story structures, wherein the method comprises: receiving, via the processing unit, output from the sensors, and adjusting unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings to maintain properties within the interior spaces and/or passageways at desired values.

(81) In certain embodiments, in the adjusting steps, the properties include temperature and relative humidity in each of the interior spaces and moisture concentration in the passageways.

(82) Embodiments of the disclosure broadly relates to methods for controlling air flow in a multi-story structures, wherein the method comprises: receiving, via the processing unit, output from the sensors, and adjusting unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and passageway air flow rates through the passageways and into the surroundings to maintain properties within the interior spaces and/or passageways at desired values.

(83) In certain embodiments, in the adjusting steps, the properties include temperature and relative humidity in each of the interior spaces and moisture concentration in the passageways.

Suitable Components for Use in the Disclosure

(84) Suitable treating compositions include, without limitation, mold control compositions, pesticides, insecticides, fungicides, bactericides, rodent poisons, other control compositions, gases, fire retardant compositions, foams, and mixtures or combinations thereof.

(85) Suitable gases include, without limitation, air, nitrogen enriched air, nitrogen (all grades), argon, carbon dioxide, haloalkanes and mixtures thereof.

(86) Suitable haloalkanes include, without limitation, iodomethane (Halon 10001); bromomethane (Halon 1001), bromochloromethane, CH_2BrCl (Halon 1011), carbon tetrachloride (Halon 104), tribromofluoromethane (Halon 1103), dichlorofluoromethane (Halon 112), bromodifluoromethane (Halon dibromodifluoromethane 1201), (Halon 1202), bromochlorodifluoromethane, CF_2ClBr (Halon 1211), dichlorodifluoromethane (Halon 122), bromotrifluoromethane, CBrF_3 (Halon 1301), tetrafluoromethane (Halon 14), 1,2-dichlorotetrafluoroethane (Halon 242), dibromotetrafluoroethane, $\text{C}_2\text{Br}_2\text{F}_4$ (Halon 2402), hexafluoroethane (Halon 2600), and mixtures thereof.

(87) Suitable foams include, without limitation, any of the gases and compositions introduced as a foam using a foaming agent or the co-introduction of a foaming agent and a gas or treating compositions. Suitable foaming agents include all surfactants and any other material that forms a foam when combined with a gas.

(88) Suitable flow control valves or dampers for use herein include, without limitation, control valves or dampers manufactured by Johnson Controls, Belimo, Honeywell, Siemens, Greenheck, KMC Controls, or other similar manufacturers.

(89) Suitable sensor for use in the disclosure include, without limitation, relative humidity sensors, moisture sensors, water sensors, physical sensors such as temperature sensors, pressure sensors, air flow sensors, etc., chemical sensors such as carbon dioxide sensors, carbon monoxide sensors, oxygen sensors, etc., microbial sensors such as mold sensors, fungus sensors, bacterial sensors, viral sensors, etc., other sensors, and combinations thereof.

(90) Suitable air flow sensors for use herein include, without limitation, E+E Elektronik.

Honeywell, Siemens, IFM Electronics, Air Logic, SMT, EGE, Delta OHM, Weber, EC, or other similar manufacturers.

(91) Suitable temperature sensors for use herein include, without limitation, Analog Devices, Mitsumi, Melexis, Environdata, Geokon, IST. SBE, Andigilog, AquiStar, Envco, STMicroelectronics, ZMD, Measurement Specialties, Microship, MicroDAQ, Smartec, Dallas Semiconductor, Slop Indicator, Honsberg. Maxim, ReGMet, GF Signet, Philips, Texas Instruments, FTDI Chip, Thermometrics, Spica Technology, or other similar manufacturers.

(92) Suitable water/moisture sensors for use herein include, without limitation, Converge, Edyn, Oval Digital, Sengenla, Galltec, E+E Elektronik, TE Connectivity, Kurabe Industrial Co, Ltd., Scale-Tron, Measurement Specialties, Michell Instruments, EMS Brno, Hoperf Electronics, GE Sensing. Hygrosens Instruments, Sensirion, Gefran, U.P.S.I., Precon, EME Systems, FTDI Chip, IST, Blue Earth, Vishay, or other similar manufacturers.

(93) Suitable chemical sensors for use herein include, without limitation, carbon dioxide sensors, carbon monoxide sensors, mold sensors, or other similar chemical and microbial sensors.

(94) Suitable physical sensors for use herein include, without limitation, pressure sensors, air compositional sensors, partial pressure sensors, or other physical sensors.

(95) Suitable processors or processing units for use herein include, without limitation, without limitation, digital processing units (DPUs), analog processing units (APUs), any other technology that can receive motion sensor output and generate command and/or control functions for objects under the control of the processing unit, or mixtures and combinations thereof.

(96) Suitable digital processing units (DPUs) include, without limitation, any digital processing unit capable of accepting input from a plurality of devices and converting at least some of the input into output designed to select and/or control attributes of one or more of the devices. Exemplary examples of such DPUs include, without limitation, microprocessor, microcontrollers, or the like manufactured by Intel, Motorola, Ericsson, HP, Samsung, Hitachi, NRC. Applied Materials, AMD, Cyrix, Sun Microsystem, Philips, National Semiconductor, Qualcomm, or any other manufacture of microprocessors or microcontrollers.

(97) Suitable analog processing units (APUs) include, without limitation, any analog processing unit capable of accepting input from a plurality of devices and converting at least some of the input into output designed to control attributes of one or more of the devices. Such analog devices are available from manufacturers such as Analog Devices Inc.

(98) Suitable interfaces and interface components for use herein include, without limitation, manufactured by Microsoft, Samsung, Apple, Linex, or other similar GUI or user interfaces.

(99) Suitable HVAC unit and associated components for use herein include, without limitation, AAON, Airedale International Air Conditioning, Amana Corporation, American Standard Companies, ASI Controls, Baxi, BDR Thermea, BELIMO Holding AG, Buffalo Forge Company, CaptiveAire Systems, Carrier Corporation, Chunlan Group, CIAT Group, Climaveneta, Coleman Heating & Air Conditioning, Comfort Systems USA, Cylon Controls Ltd., Daewoo Electronics, Daikin, Daikin Applied Americas, De Dietrich Remeha, De'Longhi, DeLclima, Electronic Temperature Instruments, Embraco, Fegyver-és Gépgyár, Fuji Electric, Fujitsu, Galanz, General Airconditioners, GlenDimplex, Goodman Global, Gree Electric, Haier, Hart & Cooley, Help-Link, Hisense, Hisense Kelon, Hitachi, J & E Hall, Jasun Filtration, Johnson Controls, Jøtul, Kenstar, KMC Controls, Lennox International, LG Electronics, MarinAire, Mirgor, Mitsubishi Electric, MPF Industry Group, Panasonic, PEC Power Electric Cooperation, Petra Engineering Industries Company, Rettig ICC, Rheem, Robert Bosch GmbH, Roberts-Gordon LLC, Sampo Corporation, Sanden Corporation, Sanhua, Sanyo, Shinco, Síragon, Stelrad, Taikisha, Tatung Company, Temperzone, Thermax, Thermo King, Toshiba, Tranc, Vaillant Group, Viega, Viessmann, Voltas, Walton Group, Wellman Group, Westaflex, Whirlpool Corporation, Worcester, Bosch Group, Wrightsoft, Yilida, York International, or other similar manufacturers.

DETAILED DESCRIPTION OF THE DRAWINGS OF THE DISCLOSURE

Prior Art

(100) Referring now to FIG. 1, a prior art structure, generally **100**, is shown to include a slab **102** situated on the ground **104** with a water resistant layer **106** on a bottom surface **108** of the slab **102**. The structure **100** also includes an outer wall **110** (shown here as a masonry and mortar wall) anchored to an inner wall **112** via ties **114**. The outer wall **110** includes weep holes **116**. The inner wall **112** includes studs **118**, a sheathing layer **120**, and a drywall layer **122** (e.g., gypsum board or plywood) and insulation.

(101) The structure **100** also includes a roof **140**, a ceiling **142** supported by framing **144**, an attic **146**, and an interior **148**. The structure **100** also includes an HVAC unit **150** situated in the attic **146**. The HVAC unit **150** includes an inlet conduit **152** having an inlet **154** for pulling air **156** from the interior **148** into the HVAC unit **150**. The HVAC unit **150** also includes an outlet conduit **158** having an outlet **160** adapted to push conditioned air **162** into the interior **148**.

(102) The structure **100** is ill equip to reduce, prevent, inhibit, or eliminate water, moisture and/or mold intrusion into exterior walls and as shown later herein, interior walls.

(103) In U.S. Pat. No. 7,247,090B2, the inventor disclosed systems and methods for reducing water and moisture accumulation in external walls by providing a path way for conditioned air pass through the external walls and exhaust out through weep holes in the external walls. The present invention describes improved systems, apparatuses, and methods for reducing, removing, inhibiting, or eliminating water and/or moisture and/or their accumulation and/or mold growth in exterior and interior wall of habitable structures.

First Embodiment

(104) Referring now to FIG. 2A, an embodiment of a structure of this disclosure, generally **200**, is shown to include a slab **202** situated on the ground **204** with a water resistant layer **206** on a bottom surface **208** of the slab **202**. It should be noted that the water resistant layer **206** may be disposed on the top surface of the slab **202** or both surfaces may include a water resistant layer **206**. The structure **200** also includes an outer wall **210** (shown here as a masonry and mortar wall) anchored to an inner wall **212** via ties **214**. The outer wall **210** includes a masonry sealer layer **216** and weep holes **218**. The inner wall **212** includes studs **220**, bracings between studs **222**, a sheathing and/or drywall layer **224**, and an optional drywall layer **226**, which may be insulating or non-insulating. The ties **214** are angled up and affixed to the studs **220**.

(105) The structure **200** also includes an insulated roof **240**, a ceiling **242** supported by framing **244**, an attic **246**, and an interior **248**. The structure **200** also include an HVAC unit **250** situated in the attic **246**. The HVAC unit **250** includes an inlet conduit **252** having an inlet **254**, wherein the inlet **254** pulls unconditioned air **256** from the surrounding **258** into the HVAC unit **250**. The HVAC unit **250** also includes an outlet conduit **260** having an outlet **262** adapted to push conditioned air **264** into the interior **248**. The structure **200** also includes an outlet conduit **266** having an inlet **268** and an outlet **270**, wherein the inlet **268** receives interior exhaust conditioned air **272** and the outlet **268** exhausts the interior exhaust conditioned air **272** into the attic **246**. The structure **200** also includes a unidirectional valve **274** so that the exhausted conditioned air **272** flows (indicated by arrows) from the attic **246** into an air flow path **276** between the outer wall **210** and the inner wall **212** and out the weep holes **218** exhausting spent conditioned air **278** into the surrounding **258**. The flow of the conditioned air from the attic through the pathways and out of the weep holes may be used to push water and/or moisture out of the walls to facilitate wall drying or to decrease/reduce or inhibit water/moisture accumulation or to decrease/reduce or inhibit mold growth in the outer wall and/or to control the moisture content in the walls.

(106) Referring now to FIG. 2B, the structure of FIG. 2A further including an attic outlet conduit **280** having an inlet **282** and an outlet **284** for exhausting the attic exhaust conditioned air **286** from the attic **246**. The attic outlet **280** may also include a control valve **288** so that a portion of the attic exhausted conditioned air **282** flows (indicated by arrows) from the attic **246** into the surrounding **258**.

Second Embodiment

(107) Referring now to FIG. 3A, another embodiment of a structure of this disclosure, generally **300**, is shown to include a slab **302** situated on the ground **304** with a water resistant layer **306** on a bottom surface **308** of the slab **302**. Again it should be noted that the water resistant layer **306** may be disposed on the top surface of the slab **302** or both surfaces may include a water resistant layer **306**. The structure **300** also includes an outer wall **310** (shown here as a masonry and mortar wall) anchored to an inner wall **312** via ties **314**. The outer wall **310** includes a masonry sealer layer **316** and weep holes **318**. The inner wall **312** includes studs **320**, bracings between studs **322**, a sheathing and/or drywall layer **324**, and an optional drywall layer **326**, which may be insulating or non-insulating. The ties **314** are angled up and affixed to the studs **320**.

(108) The structure **300** also includes an insulated roof **340**, a ceiling **342** supported by framing **344**, an attic **346**, and an interior **348**. The structure **300** also includes an HVAC unit **350** situated in the attic **346**. The HVAC unit **350** includes an inlet conduit **352** having an inlet **354**, wherein the inlet **354** pulls unconditioned air **356** from the surrounding **358** into the HVAC unit **350**. The HVAC unit **350** also includes an outlet conduit **360** having an outlet **362** adapted to push conditioned air **364** into the interior **348**. The structure **300** also includes an outlet conduit **366** having an inlet **368**, an outlet **370**, and a control valve **371**, wherein the inlet **368** receives interior exhaust conditioned air **372** and the outlet **370** exhausts the interior exhaust conditioned air **372** into the surrounding **358** and wherein the control valve **371** controls the amount of air being exhausted through the outlet **370**. The structure **300** may also include a control valve **374** so that the exhausted conditioned air **370** flows (indicated by arrows) from the attic **346** into an air flow path or passageway **376** between the outer wall **310** and the inner wall **312** and out the weep holes **318** exhausting spent conditioned air **378** into the surrounding **358**. However, in certain embodiments, the control valve **371** will be sufficient to direct the desired amount of conditioned air to the passageways, unless there are more than one passageways, then each passageway would include a control valve **371** that would control the air flow into each of the passageways/pathways. The flow of the conditioned air from the attic through the pathways or passageways and out of the weep holes may be used to push water and/or moisture out of the walls to facilitate wall drying or to decrease/reduce or inhibit water/moisture accumulation or to decrease/reduce or inhibit mold growth in the outer wall and/or to control the moisture content in the walls.

(109) Referring now to FIG. 3B, the structure of FIG. 3A further including an attic outlet conduit **380** having an inlet **382** and an outlet **384** for exhausting the attic exhaust conditioned air **386** from the attic **346**. The attic outlet **380** may also include a control valve **388** so that a portion of the attic exhausted conditioned air **382** flows (indicated by arrows) from the attic **346** into the surrounding **358**.

(110) Expanded Views of Walls

(111) Referring now to FIGS. 4A&B, expanded views of embodiments of an exterior wall of a structure of this disclosure, generally **400**, are shown. FIG. 4A depicts an expanded cross-sectional view of the walls, while FIG. 4B depicts successive cut away elevation view of the exterior wall.

(112) Looking at FIG. 4A, the exterior wall **400** includes an outer wall assembly **402** comprising an outer wall **404** and a sealer layer **406** and an inner wall assembly **408** comprising studs **410**, bracings between studs **412**, and insulation **414** (shown only in FIG. 4B), an thermal and/or moisture barrier layer **416**, and sheathing and/or drywall layer **418**, which may be insulating or non-insulating. The exterior wall **400** also includes ties **420** anchoring the outer wall **404** to the studs **410** or the bracings between studs **412** of the inner wall assembly **408** and an air pathway or passageway **422** disposed between the outer wall assembly **402** and the inner wall assembly **408**.

(113) Looking at FIG. 4B, the wall **400** depicts successive cutaways showing each layer of the wall **400**. The wall **400** includes the outer wall **404**, then the sealer layer **406**, then the air pathway **422**, and then the studs **410**, the bracings between studs **412**, and the insulation **414**. The wall **400** then include the barrier layer **416**, and then the layer **418**.

Third Embodiment

(114) Referring now to FIG. 5A, another embodiment of a structure of this disclosure, generally **500**, is shown to include an outer wall **502**, an inner wall **504**, an air flow control valve **506**, an air flow pathway **508** interposed between the outer wall **502** and the inner wall **504**, a roof **510**, an attic **512**, and a slab **514**. The structure **500** includes a ceiling **516** and support framing **518**.

(115) The structure **500** also includes an air circulation assembly **520** including an HVAC unit **522** including an air intake conduit **524** having an inlet **526** and a control valve **527** for supplying unconditioned air **528** from an outside **530** to the HVAC unit **522** and an attic air conduit **532** having an air control valve **534** and an inlet **536** for supplying exhaust conditioned air **538** to the HVAC unit **522** from the attic **512**. The circulation assembly **520** also includes an outlet conduit **540** having a control valve **542** and an outlet **544** for sending conditioned air **546** into an interior **548** of the structure **500**. It should be recognized that the structure **500** may include a plurality of outlets **544** associated with the outlet conduit **540** for sending the conditioned air **546** into the interior **548**. The circulation assembly **520** also includes an exhaust conduit **550** having an inlet **552** that withdraws exhaust conditioned air **554** from the interior **548** and an outlet **556** through which the exhaust conditioned air **554** is discharged into the attic **512**. Some of the attic air **554** then exits through the control valve **506** and through the air flow pathway **508** shown by the dark grey arrows and out through weep holes or channels **558** into the outside **530**.

(116) Referring now to FIG. 5B, the structure of FIG. 5A further including an attic outlet conduit **560** having an inlet **562** and an outlet **564** for exhausting the attic exhaust conditioned air **566** from the attic **512**. The attic outlet **560** may also include a control valve **568** so that a portion of the attic exhausted conditioned air **566** flows (indicated by arrows) from the attic **512** into the surrounding **530**.

Fourth Embodiment

(117) Referring now to FIG. 5C, another embodiment of a structure of this disclosure, generally **500**, is shown to include an outer wall **502**, an inner wall **504**, an air flow control valve **506**, an air flow pathway **508** interposed between the outer wall **502** and the inner wall **504**, a roof **510**, an attic **512**, and a floor **514**. The structure **500** includes a ceiling **516** and support framing **518**.

(118) The structure **500** also includes an air circulation assembly **520** including an HVAC unit **522** including an air intake conduit **524** having an inlet **526** and a control valve **527** for supplying unconditioned air **528** from an outside **530** to the HVAC unit **522** and an attic air conduit **532** having an air control valve **534** and an inlet **536** for supplying exhaust conditioned air **538** to the HVAC unit **522** from the attic **512**. The circulation assembly **520** also includes an outlet conduit **540** having a control valve **542** and an outlet **544** for sending conditioned air **546** into an interior **548** of the structure **500**. It should be recognized that the structure **500** may include a plurality of conduits **540**, control valves **542**, and outlets **544** for sending the conditioned air **546** into the interior **548**. The circulation assembly **520** also includes an exhaust conduit **550** having an inlet **552** that withdraws exhaust conditioned air **554** from the interior **548** and an outlet **556** through which the exhaust conditioned air **554** is discharged into the attic **512**. It should be recognized that the structure **500** may also include a plurality of conduits **550**, inlets **552**, and outlets **556** for withdrawing exhaust error **554** from the interior **548**. The system **520** also includes a wall conditioned air conduit **560** having a control valve **562** through which conditioned air shown by the dark grey arrows is discharged into the air flow pathway **508** via an outlet **564** into the pathway **508**, through the pathway **508**, and out through weep holes or channels **558** into the outside **530**. Again, it should be recognized that the structure **500** may also include a plurality of conduits **560**, valves **562**, and outlets **564** associated with a plurality of pathways **508** associated with exterior or interior walls.

(119) Referring now to FIG. 5B, the structure of FIG. 5A further including an attic outlet conduit **566** having an inlet **568** and an outlet **570** for exhausting the attic exhaust conditioned air **572** from the attic **512**. The attic outlet **560** may also include a control valve **574** so that a portion of the attic

exhausted conditioned air **566** flows (indicated by arrows) from the attic **512** into the surrounding **530**.

(120) First Multistory Building Embodiment

(121) Referring now to FIG. **6A**, an embodiment of a multistory structure of this disclosure, generally **600**, is shown to include a lower floor **602**, outer/exterior wall or walls **604**, inner walls **606**, and air flow pathways **608** interposed between the outer wall or walls **604** and the inner walls **606**, wherein the pathways or passageways **608** include inlets **610** having control valves **612** and outlet channels **614**. The structure **600** also includes floors **616**, ceilings **618**, dead spaces **620** interposed between the intermediate floors **616** and the ceiling **618**, and a roof **622**. It should be recognized that the outer walls **604** may be transparent (made of glass) and the air flow passageways **608** comprise a gap between two transparent layer or panes of glass through which air may flow.

(122) The structure **600** also includes an air circulation assembly or system **630** including an HVAC unit **632** including an unconditioned air intake conduit **634** having an inlet **636** and a control valve **638** for supplying unconditioned air **640** from an outside **642** to the HVAC unit **632** and optionally an exhaust conditioned air intake conduit **644** having an inlet **646** and a control valve **648** for supplying exhaust conditioned air **670** from the dead spaces **620** to the HVAC unit **632** via the conduits **644** and **634**.

(123) The system **630** also includes a main conditioned air conduit **652** from which floor conditioned air conduits **654** branch. Each of the floor conduits **654** include a flow control valve **656** and terminate in an outlet **658** passing through their respective ceilings **618** (only one per floor shown here, but of course, each floor will generally have a plurality of outlets **658**). The floor conduits **654** discharge conditioned air **660** into interiors **662** between the floors **602** and **616** and the ceilings **618**.

(124) The system **630** also includes floor outlet conduits **664** having inlets **666** and outlets **668** for discharging exhaust conditioned air **670** into the dead spaces **620**, which then progresses through the control valves **612** through the pathways **608** and out of the channels **614** into the outside **642** or through the conduits **644** and **634** to the HVAC unit **632**.

(125) The system **630** also includes a dead space return conduit **672** configured to allow exhaust conditioned air to flow from each dead space **620** with inlets **674** and an outlet **676** in the last floor dead space, wherein the return conduit **672** is configured to supply return exhausted conditioned air to the HVAC unit via the conduit **644**. The return conduit **672** may include a control valve at each floor (not shown) to control the amount of air introduced into the return conduit **672** from each of the dead spaces **620**.

(126) Second Multistory Building Embodiment

(127) Referring now to FIG. **6B**, another embodiment of a multistory structure of this disclosure, generally **600**, is shown to include a lower floor **602**, outer/exterior wall or walls **604**, inner walls **606** and air flow pathways **608** interposed between the outer wall or walls **604** and the inner walls **606** and exiting through exit channels **610**. The structure **600** also includes floors **616**, ceilings **618**, dead spaces **620** interposed between the intermediate floors **610** and the ceiling **612**, and a roof **622**. Again, it should be recognized that the outer walls **604** may be transparent (made of glass) and the air flow passageways **608** comprise a gap between two transparent layer or panes of glass through which air may flow.

(128) The structure **600** also includes an air circulation assembly **630** including an HVAC unit **632** including an unconditioned air intake conduit **634** having an inlet **636** and a control valve **638** for supplying unconditioned air **640** from an outside **642** to the HVAC unit **632** and optionally an exhaust conditioned air intake conduit **644** having an inlet **646** and a control valve **648** for supplying exhaust conditioned air **670** from the dead spaces **620** to the HVAC unit **632** via the conduit **634**.

(129) The system **630** also include a main conditioned air conduit **652** from which floor

conditioned air conduits **654** branch. Each of the floor conduits **654** include flow controllers **656** and terminate in outlets **658** passing through their respective ceilings **618** (only one per floor shown here, but of course, each floor will generally have a plurality of outlets **658**). The floor conduits **654** discharge conditioned air **660** into interiors **662** between the floors **616** and the ceilings **618**. The system **630** also includes pathway floor outlet conduits **664** having inlets **666** and outlets **668** for discharging exhaust conditioned air **670** into the dead spaces **620**.

(130) The system **630** also includes conditioned air pathway conduits **672** including a control valves **674** and outlets **676** through which conditioned air shown in the dark gray arrows flows into the pathways **608** and then out through the channels **610**.

(131) The system **630** may also include an exhaust conduit **678** having an outlet **680** and a control valve **682** for exhausting the air **670** from the dead spaces **620** to the outside **642**.

(132) Third Multistory Building Embodiment

(133) Referring now to FIG. **6C**, another embodiment of a multistory structure of this disclosure, generally **600**, is shown to include in addition to the other features disclosed above, exhaust channels **610** at each floor. Of course, the exhaust channels may be also include control valves to control the rate of air being exhausted from the passageways. Additionally, the passageways may extend between floors so that the flow rate of exhausted air may be controlled floor by floor. It should be recognized that all of the multi-story structures will include a return system for returning exhaust conditioned air to the HVAC unit. Additionally, all structures may include an air circulation systems that monitors carbon dioxide, carbon monoxide, oxygen levels, humidity, and temperature, and to control the air flow into and out of the interior spaces and into and out of the passageways to maintain these properties at desirable levels.

(134) Pier and Beam Building Embodiment

(135) Referring now to FIGS. **7A&B**, embodiment of a pier and beam structure of this disclosure, generally **700**, is shown to include a pier and beam facade **702** having air vents **704** and solid portions **706** and piers **708** extending from the ground or below the ground **710** to a floor **712** and a dead space **714** between the floor **712** and the ground **710**. The structure **700** also includes outer walls **716**, inner walls **718**, and air flow pathways or passageways **720**. The structure **700** also includes interiors **722**. It should be recognized that the pier and beam structure **700** may include any or all of the circulation systems, control systems, and sensor systems set forth herein. It should be recognized that there does not have to be an air vent **704** between every pair of piers **708** and the number of air vents **704** are generally a matter of design.

(136) Embodiments Wall Air Distribution Systems

(137) Referring now to FIGS. **8A-D**, four embodiments of wall air distribution systems of this disclosure, generally **800**, **830**, **850**, and **870**, respectfully, are shown.

(138) Looking at FIG. **8A**, the wall air distribution system **800** includes an inlet **802**, an air distribution conduit **804** having an outlet **806** that passes through a ceiling **808** and ends within an air pathway **810** disposed between an outer wall section and an inner wall section (not shown) of an interior or exterior wall **812**. Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlet **802** through the distribution conduit **804** and out the outlet **806** depicted by the dark gray arrows and then through the pathway **810** and out through weep holes or outlet channels (not shown), but fully described herein.

(139) Looking at FIG. **8B**, the wall air distribution system **830** includes a plurality of inlets **832**, a distribution conduit **834**, and a plurality of outlets **836**. Each of the outlets **836** passes through a ceiling **838** and ends within one of a plurality of air pathways **840** formed by a plurality of pathway dividers **842** within an interior or exterior wall **844**. The pathways **840** are disposed between an outer wall section and an inner wall section (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlets **832** through the distribution conduit **834** and out the outlets **836** and then through the pathways **840** as depicted by the dark gray arrows and out through weep holes or outlet channels (not shown).

(140) Looking at FIG. 8C, the wall air distribution system **850** includes an inlet **852**, an air distribution conduit **854** having a plurality of flow control valves **856** and a plurality of outlets **858**. Each of the outlets **858** passes through a ceiling **860** and ends within a plurality of air pathways **862** formed by a plurality of pathway dividers **864** within an interior or exterior wall **866**. The pathways **862** are disposed between an outer wall section and an inner wall section (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlet **852** through the distribution conduit **854** and out the outlets **858** as depicted by the dark gray arrows and then through the pathways **862** and out through weep holes (not shown), where the air flow through each of the pathways **862** is controlled by the control valves **856**.

(141) Looking at FIG. 8D, the wall air distribution system **870** includes a plurality of inlets **872**, a plurality of distribution conduits **874**, a plurality of flow control valves **876**, and a plurality of outlets **878**. Each of the outlets **878** passes through a ceiling **880** and ends within one of a plurality of air pathways **882** formed by a plurality of pathway dividers **884** within an interior or exterior wall **886**. The pathways **882** are disposed between an outer wall and an inner wall of an interior or exterior wall (not shown). Conditioned air proceeds from an HVAC unit (not shown) and flows into the inlets **872** through the distribution conduits **874** and out the outlets **878** depicted by the dark gray arrows and then through the pathways **882** and out through weep holes (not shown), where the air flow through each of the pathways **882** is controlled by the control valves **876**.

(142) Embodiment Wall Air Distribution Systems

(143) Referring now to FIG. 9A, a structure, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and an air flow pathway **912** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**.

(144) The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior (not shown) of the structure **900**, while the secondary conduit **938** supplies conditioned air to the air flow pathway **912** so that conditioned air may flow down the pathway **912** pushing out water and/or moisture through weep holes or drainage channels and drying the outer wall **902** and/or the inner wall **904**. Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein. The assembly **920** also includes an outlet **946** associated with an outlet conduit (not shown) for exhausting air from the interior of the structure **900** to the surroundings or to an attic or similar space associated with the structure **900**.

(145) Looking at FIG. 9B, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and air flow pathways **912a-d** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**. The pathways **912** are isolated by corner dividers **916** that prevent air exchange between the pathways **912a-d**.

(146) The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior of the structure **900** (not shown). The assembly **920** also includes an outlet **946** associated with an outlet conduit (not shown) for withdrawing air from the interior of the structure and exhausting either to the surroundings or to an attic or similar space associated with the structure **900**.

(147) The secondary conduit **938** terminated in a distribution unit **948** that supplies conditioned air to four wall conduits **950a-d**, where the conduits **950a-d** include flow control valves **952a-d**. The four conduits **950a-d** supply conditioned air to the four pathways **912a-d** so that the conditioned air may flow down the pathway **912a-d** pushing out water and moisture through weep holes or drainage channels and drying the outer wall **902** and the inner wall **904**. The control valve **952a-d** controls the amount of conditioned air that flows into each pathway **912a-d** so that each wall section may be dehydrated at different rates depending on water, moisture and/or mold content.

(148) Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein.

(149) Looking at FIG. **9C**, generally **900**, is shown to include an outer wall **902** and an inner wall **904** having a thermal and/or moisture barrier layer **906**, studs **908**, a sheathing and/or drywall layer **910**, which may be insulating or non-insulating, and sixteen air flow pathways **912a-p** between an inner surface of the outer wall **902** and the barrier layer **906**. The structure **900** also includes a ceiling **914** upon which is disposed an air conditioning and distribution assembly **920**. The pathways **912a-p** are isolated by dividers **916** that prevent air exchange between the pathways **912a-p**.

(150) The assembly **920** includes an air inlet **922**, an air inlet conduit **924** connected to an HVAC unit **928**. The HVAC unit **928** includes an outlet **930** and an outlet conduit **932**. The outlet conduit **932** splits into a main conduit **934** having a main control valve **936** and a secondary conduit **938** having a secondary control valve **940**. The main conduit **934** supplies conditioned air to outlets **942** associated with registers **944** for heating or cooling an interior of the structure **900** (not shown). The assembly **920** also includes an outlet **946** associated with an outlet conduit (not shown) for withdrawing air from the interior of the structure and exhausting either to the surroundings or to an attic or similar space associated with the structure **900**.

(151) The secondary conduit **938** terminated in a distribution unit **948** that supplies conditioned air to four main wall conduits **950a-d**. The four main wall conduits **950a-d** end in sixteen flow control valves **952a-p** and feed conditioned air to sixteen air distribution conduits **954a-p** that in turn supply conditioned air to the sixteen pathways **912a-p** so that the conditioned air may flow down the pathway **912a-d** pushing out water and moisture through weep holes or drainage channels and drying the outer wall **902** and the inner wall **904**. The control valve **952a-p** controls the amount of conditioned air that flows into each pathway **912a-p** so that each wall section may be dehydrated at different rates depending on water, moisture and/or mold content.

(152) Of course, the same wall assembly may be used to dry out interior walls, but with different drainage and exhaust channels as described herein.

Fifth Embodiment

(153) Referring now to FIG. **10A**, an embodiment of a structure of this disclosure, generally **1000**, is shown to include a slab **1002** situated on the ground **1004**. The structure **1000** also includes an outer wall **1006** and an inner wall **1008**. The outer wall **1006** and the inner wall **1008** may be of any type or construction. The structure **1000** also includes an outside HVAC unit **1010** disposed on an HVAC slab **1012** and including an air intake **1014** having a screen **1016** for pulling air **1018** from the surroundings into the HVAC unit **1010**. The structure **1000** also includes an outlet **1020** and an outlet conduit **1022** leading through a roof **1024**, through an attic space **1026**, and through a ceiling **1028** supported by framing **1030** to an interior inlet **1032** from which conditioned air **1034** is discharged into an interior **1036** of the structure **1000**. The structure **1000** also includes an interior outlet **1038** associated with an outlet conduit **1040** having an exhaust outlet **1042** for withdrawing air from the interior **1036** and discharging exhaust air **1044** into the attic **1026** or surrounding. The HVAC unit **1010** also includes a conditioned air wall discharge conduit **1046** including a control valve **1048** for controlling a rate and amount of conditioned air discharged into a wall pathway **1050** flows through the pathway **1048** and out exhaust holes **1052**, which may be weep holes if the outer wall is masonry and mortar or just discharge channels if the outer wall is of some other

construction. It should be recognized that in multistory building, the space into which the air is exhausted and that may be used to dry out walls may be any dead space including the space between floors or any other dead space that would allow air to flow through the walls and out the outlet channels. Of course, it should also be recognized that some structures the air flow may be from the bottom of the wall up especially in multistory buildings.

(154) Referring now to FIG. 10B, the structure of FIG. 10A further including an attic outlet conduit **1054** having an inlet **1056** and an outlet **1058** for exhausting the attic exhaust conditioned air **1060** from the attic **1026**. The attic outlet conduit **1054** may also include a control valve **1062** so that a portion of the attic exhausted conditioned air **1060** flows (indicated by arrows) from the attic **1026** into the outside.

(155) Expanded Views of Exterior Walls with Sensors

(156) Referring now to FIG. 11A, an expanded view of an embodiment of an outer wall assembly of a structure of this disclosure, generally **1100**, is shown to an outer wall **1102** and a scaler layer **1104** and an inner wall assembly **1106** including studs **1108**, an moisture and/or vapor barrier layer **1110**, an insulating sheathing and/or drywall layer **1112** situated on a slab **1114**. The wall assembly **1100** also includes ties **1116** anchoring the outer wall **1102** to the studs **1108** of the inner wall assembly **1106**. The wall assembly **1100** also includes an air pathway **1118** disposed between the outer wall **1102** and the barrier layer **1110**. The wall assembly **1100** also includes exhaust ports, holes, or channels **1120** through which conditioned air is exits the pathway **1118**.

(157) The wall assembly **1100** also includes wall temperature sensors **1122**, moisture/water sensors **1124** and an air flow rate sensor **1126**. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

(158) Looking at FIG. 11B, an expanded view of another embodiment of an outer wall assembly of a structure of this disclosure, generally **1150**, is shown to an outer wall **1152** and a sealer layer **1154** and an inner wall assembly **1156** including studs **1158**, an moisture and/or vapor barrier layer **1160**, an insulating sheathing and/or drywall layer **1162** situated on a slab **1164**. The wall assembly **1150** also includes ties **1166** anchoring the outer wall **1152** to the studs **1158** of the inner wall assembly **1156**. The wall assembly **1150** also includes an air pathway **1168** disposed between the outer wall **1152** and the barrier layer **1160**. The wall assembly **1150** also includes exhaust ports, holes, or channels **1170** through which conditioned air is exits the pathway **1168**. The wall assembly **1156** also includes an wall air conduit **1172** for supplying conditioned air to the pathway **1168** and a treating solution reservoir **1174** disposed on a beam **1176**, a treating solution treating dispensing unit **1178**, a dispensing conduit **1180** ending in a nozzle **1182** for spraying the treating solution **1184** into the pathway **1168**.

(159) The wall assembly **1100** also includes wall temperature sensors **1186**, moisture/water sensors **1188** and an air flow rate sensor **1190**. The sensors and control valves are all controlled by a controller describe more fully in FIG. 13.

(160) The wall assembly **1100** may also include a control valve **1192** associated with the channels **1170** so that the passageways or pathways **1168** may be pressurized or cut off from air flow. Thus, the passageways or pathways **1168** may be flooded with a fire retardant gas or foam and pressurized to help impede the spread of a fire.

(161) Expanded Views of Interior Walls with Sensors and Dispensing Assemblies

(162) Referring now to FIG. 12A, an expanded view of an embodiment of an inner wall assembly of a structure of this disclosure, generally **1200**, is shown to include studs **1202**, a first insulating sheathing and/or drywall layer **1204** and a second insulating sheathing and/or drywall layer **1206**. The inner wall assembly **1200** also includes an air pathway **1208** disposed in a center **1210** of the assembly **1200**. The assembly **1200** also includes a conditioner air conduit **1212** that supplies conditioned air into the pathway **1208** and an exhaust conduit **1214** that exhaust air, water, and/or moisture from the interior of the wall assembly **1200** disposed in a sub floor **1216** supported by support members **1218** on a slab **1220** and is vented out of the structure **1200** via an outlet **1222**,

where conditioned air flow is shown by the dark gray arrows. Of course, the pathway **1208** may require holes or other opening to be provided through the inner wall cross members (not shown) so that the conditioned is capable of reducing, decreasing, and/or eliminating water, moisture or mold accumulation in the wall assembly **1200** or facilitate drying the wall assembly **1200** after a water, moisture or mold event.

(163) The wall assembly **1200** also includes wall temperature sensors **1224**, moisture/water sensors **1226**, and an air flow rate sensor **1228**. The sensors and control valves are all controlled by a controller describe more fully in FIG. **13**.

(164) Looking at FIG. **12B**, an expanded view of another embodiment of an inner wall assembly of a structure of this disclosure, generally **1250**, is shown to include studs **1252**, a first insulating sheathing and/or drywall layer **1254** and a second insulating sheathing and/or drywall layer **1256**. The inner wall assembly **1250** also includes an air pathway **1258** disposed in a center **1260** of the assembly **1250**. The assembly **1250** also includes a conditioner air conduit **1262** that supplies conditioned air into the pathway **1258** and an exhaust conduit **1264** that exhaust air, water, and/or moisture from the interior of the wall assembly **1250** disposed in a sub floor **1266** supported by support members **1268** on a slab **1270** and is vented out of the structure **1250** via an outlet **1272**, where conditioned air flow is shown by the dark gray arrows. The assembly **1250** also includes a treating solution dispensing conduit **1274** ending in a nozzle **1276** for spraying a treating solution **1278** into the pathway **1258**. The treating solution reservoir and dispensing mechanism are not shown, but may be as shown in FIG. **11B** or of any other form used in the art to dispense a liquid, gas, or foam treating composition.

(165) Of course, the pathway **1258** may require holes or other opening to be provided through the inner wall cross members (not shown) so that the conditioned air is capable of reducing, decreasing, and/or eliminating water, moisture or mold accumulation in the wall assembly **1250** or facilitate drying the wall assembly **1250** after a water, moisture or mold event.

(166) The wall assembly **1250** also includes wall temperature sensors **1280**, moisture/water sensors **1282**, and an air flow rate sensor **1284**. The sensors and control valves are all controlled by a controller describe more fully in FIG. **13**.

(167) The wall assembly **1200** may also include a control valve **1286** associated with the exhaust conduit **1264** so that the passageways or pathways **1258** may be pressurized or cut off from air flow. Thus, the passageways or pathways **1258** may be flooded with a fire retardant gas and pressurized to help impede the fire.

(168) While FIGS. **11B** and **12B** demonstrate that introduction of compositions into the passageways to control pests, insects, molds, fungi, bacteria, and/or other pests, the compositions being dispensed may be fire suppressants and/or retardants that may be used to retard or impede the progression of fire through a structure.

(169) Control System Embodiment

(170) Referring now to FIG. **13**, an embodiment of a control system, generally **1300**, of a structure of this disclosure is shown to include a power supply **1302** (which may of course be the a power outlet in the structure or a dedicated power supply or power conditioning unit), a processor or processing unit **1304**, one or a plurality of user interface or panel units **1306** including touch screen **1308** (or other interactive user interface), and optionally one or a plurality of remote control units **1310** (which may be a cell phone, a lap top, or desk top computer). The interface or panel units **1306** may be associated with each interior space or a plurality of interior spaces in communication with the processor **1304**. By setting air properties in each of the interior spaces via the panels **1306**, the air properties in each of the interior spaces or coupled interior spaces may be separately and independently controlled by the processor **1304**. Additionally, air properties in each of the passageways may be entered into the panels **1306** communicated to the processor **1304** so that the air properties in each passageway may be independently and separately controlled. Of course, the processor **1304** may be programmed so that air properties in each of the interior spaces may be

collectively controlled based on the number of people and heat generating equipment in each of the interior spaces. The processor **1304** include memory, communication hardware and software, and firmware or other software for controlling HVAC units, sensors, control valves, and dispensing units that dispense treating solutions. It should be recognized that passive air control valves are not be controlled by the processing unit **1304**, but may be controlled by the air flow.

(171) The power supply **1302** supplies power to the processor **1304**, the interface **1306**, an HVAC unit **1312**, control valves **1316**, temperature sensors **1318**, water/moisture sensors **1320**, air flow sensors **1322**, and treating units **1324** via power supply wires **1326**. While I have disclosed using temperature sensors, water/moisture sensors, and air flow sensors, the systems of this invention may also include other sensors such as chemical sensors **1328** (e.g., carbon dioxide, carbon monoxide, mold, smoke, etc.) or other physical sensors **1330** (e.g., pressure, etc.) and any other sensors **1332** that may assist in reducing, removing, inhibiting and/or eliminating water, moisture, and/or mold or drying walls exposed to a water, moisture, and/or mold event.

(172) The processor **1304** is in two way communication with the interface **1306**, the remote control **1310**, the HVAC unit **1312**, the sensors **1318**, **1320**, **1322**, **1328**, **1330**, and **1332**, and the treating units **1324** via communication pathways **1334** (shown here as wires for the interface **1306**, the HVAC unit **1312**, the blower **1314**, the sensors **1318**, **1320**, **1322**, **1328**, **1330**, and **1332**, and the treating units **1320** or a communication pathway **1336** (shown here as wireless for the remote control **1310**). Of course, all of the communication pathways **1324** may also be wireless or a combination of wired and wireless communication pathways. Additionally, the interface **1306**, the remote control **1310**, the HVAC unit **1312**, the blower **1314**, the sensors **1316** and **1318** all include communication hardware and software for communication with the processor **1304**, whether via wired or wireless communication pathways. Additionally, the remote control unit **1310** includes its own power supply, memory, processor, interface, touch screen or the like and human cognizable output devices such as speakers.

(173) Window Systems

(174) Referring now to FIGS. **14A-C**, three views of an embodiment of a window system, generally **1400**, are shown. Looking at FIG. **14A**, the window system **1400** includes an exterior wall assembly **1410**, an interior wall assembly **1430**, and a window assembly **1450**.

(175) The exterior wall assembly **1410** includes an exterior wall **1412** having an interior **1414**, an outer surface **1416**, and an inner surface **1418**. The exterior wall assembly **1410** may also include a moisture and/or air barrier layer **1420** having an inner surface **1422** and an outer surface **1424**. The exterior wall **1412** may be constructed out of brick, stucco, planks, any other exterior building material and mixtures and combinations thereof.

(176) interior wall assembly **1430** includes an air/moisture barrier layer **1432** having an inner surface **1434** an outer surface **1436**. The interior wall assembly **1430** also includes a sheating layer **1438** having an inner surface **1440** and an outer surface **1442**. The interior wall assembly **1430** also includes an interior **1444**. The interior **1444** includes an insulation layer **1446** having an inner surface **1448** and an outer surface **1450**, and studs **1452**. The interior wall assembly **1430** also includes an interior wall **1454** having an inner surface **1456** an outer surface **1458**. The interior wall assembly **1430** also includes a steel angle member **1460** and a wall air flow pathway **1462**. The air flow pathway **1462** is disposed between the inner surface **1418** of the exterior wall **1412** or an inner surface **1422** of the optional moisture and/or air barrier layer **1420**.

(177) The window assembly **1470** includes shim members **1472**, a window frame member **1474**, a wire mesh member **1476**, and a window sill **1478**. The window assembly **1470** also includes an outer single, double, or triple pane window **1480** having an inner surface **1482** and an outer surface **1484**. The window assembly **1470** also includes an interior single, double, or triple pane window **1486** having an inner surface **1488** and an outer surface **1490**. The window assembly **1470** also includes a window air flow pathway **1492** interposed between the outer surface **1484** of the outer window **1480** and an inner surface **1488** of the inner window **1486**. The wall air flow pathway **1462**

is connected to the window air flow pathway **1466**. These two air flow pathways **1462** and **1492** provide cooling or heating to the walls and windows and provide dry air to remove water, moisture, and/or contaminants in the walls or windows.

Experiments of the Disclosure

(178) Buildings are a necessary part of our society to provide a safe, habitable, and protected environment. Typically, the more safe, habitable, and protected the environment; the higher the cost to operate the building. Outside air necessary to diminish indoor air contamination (COVID-19, mold, radon, etc.) results in increased energy costs. Structural Ventilation of the Building Envelope is a system that introduces continuously conditioned positively pressured outdoor air into the interior habitable spaces of a structure to be exhausted into the building envelope and subsequently to the exterior. Heat gain and loss through the building envelope are significantly reduced. Multiple studies of existing buildings have been performed that indicate that moisture accumulation and damage, air contamination, and inadequate ventilation simultaneously occur in whole or in part. WUFI analyzes the hygrothermal properties of the building envelope. Analyses indicate that the reduced moisture and regulated temperatures within the building envelope, transmitted by the supply of clean outside air into the living space of the structure, diminishes mold growth and structural deterioration of the building envelope and provides significant hygrothermal resistances of the building envelope resulting in reduced HVAC equipment sizing and operating costs. Structural Ventilation of the Building Envelope is applicable to residential, institutional, medical, retail, and commercial buildings; either new construction or retrofit. Installation is performed by the current building construction workforce with currently available materials; which result in lower installation costs. Structural Ventilation of the Building Envelope is the next generation of building ventilation. A simple solution for a complex society.

(179) The Issues

(180) In the mid 1970's, the oil embargo resulted in rising energy costs. In an attempt to reduce these costs, the minimum building outside air intake was reduced to 2.5 L/s (5 cfm) per person. This minimum still exists today. The recent onset of COVID-19 brought the realization that more fresh outside air was necessary to dilute and remove the indoor air contamination. The EPA and others have stated this recommendation. Sick Building Syndrome, also identified as Building Related Illness, a condition where individuals exhibit sickness when occupying a particular building, has been linked to airborne mold contamination. Other airborne contaminants are also blamed. More fresh outside air, necessary to dilute and remove the indoor air contamination, as well as moisture and humidity control, are required. The EPA and others have stated this recommendation. Radon Contamination, prevalent over more than half of the United States, is the leading cause of lung cancer following tobacco smoke. Radon is an odorless radioactive gas that is emitted from the earth and collects within structures. The best solutions are to ventilate beneath the building, if accessible, and to increase fresh outside air intake to dilute and remove the indoor air contamination. The EPA and others have stated this recommendation. Clearly, proper ventilation is the solution.

(181) The Conflict

(182) Buildings are a necessary part of our society to provide a safe, habitable, and protected environment. Typically, the more safe, habitable, and protected the environment; the higher the cost to operate the building. The conflict arises between this environmental goal and the costs necessary to maintain this goal. One viewpoint is "The best solution to breathe contaminant free indoor air is to inject clean outdoor air into the building to dilute the contamination. The worst solution is to recirculate and filter indoor air". The opposing viewpoint is "The worst solution to minimize HVAC operating costs is to inject clean outdoor air into the building to dilute the contamination. The best solution is to recirculate and filter indoor air". Structural Ventilation of the Building Envelope simultaneously accomplishes both goals by minimizing indoor air contamination and reducing energy costs.

(183) Conventional Systems

(184) The majority of buildings are temperature controlled and humidity adjusted insulated enclosures. Referring to FIG. 15A a typical example of a HVAC system recirculates air within the building, air is drawn from within the space, conditioned, and circulated back into the space. The attic and cavity walls are unconditioned but ventilated with outdoor air. Interior air is temperature adjusted through recirculation but humidity is inadequately maintained. Outside air is introduced through infiltration through doors, windows, the perimeter walls, and attic with variable results. The building envelope collects moisture and offers conventional thermal resistance.

(185) The building, generally 1500 of FIG. 15A is shown to include a roof 1502, a brick veneer 1504, brick ties 1506, and weep holes 1508. The building 1500 is situated on a ground surface 1510 and supported by a foundation 1512. The building 1500 also include an attic 1514, a HVAC system 1516, attic insulation 1518, gypsum wall board 1520, wall insulation 1522, and exterior sheathing 1524. Unconditioned air enters through soffit vents, ridge vents, and/or the weep holes 1508. The HVAC system 1516 cools the unconditioned air and recirculates it back into an interior of the building 1526 through vents not shown here. Additionally, unconditioned air enters the interior 1526 via infiltration through the attic insulation 1518 and wall insulation 1522.

(186) Some buildings have heat exchangers, as shown in FIG. 15B, where some fresh outside air is introduced into the HVAC system and some indoor air is exhausted out of the building through the HVAC system. The heat exchanger works with the passage of outdoor air next to indoor air, allowing the outdoor air to be tempered closer to the indoor air temperature.

(187) The building, generally 1550 of FIG. 15B is shown to include a roof 1552, a brick veneer 1554, brick ties 1556, and weep holes 1558. The building 1550 is situated on a ground surface 1560 and supported by a foundation 1562. The building 1550 also include an attic 1564, a HVAC/heat exchange system 1566, attic insulation 1568, gypsum wall board 1570, wall insulation 1572, and exterior sheathing 1574. Unconditioned air enters through soffit vents, ridge vents, and/or the weep holes 1558. The HVAC system 1566 cools the unconditioned air and recirculates it back into an interior of the building 1576 through vents not shown here. Additionally, unconditioned air enters the interior 1576 via infiltration through the attic insulation 1568 and wall insulation 1572.

(188) Other than small energy savings from the heat exchanger, the building characteristics are identical to FIG. 15A. This leads us to the need for Structural Ventilation (SV) of the Building Envelope.

(189) Structural Ventilation of the Building Envelope

(190) The system was developed in the early 2000's and patented in 2007 and 2022. The system development began with the aid of the MOIST software and elaborate spreadsheets filled with hygrothermal equations. Technology has since improved. Structural Ventilation of the Building Envelope (SV) is a system that introduces continuously conditioned, positively pressured, outdoor air into the interior habitable spaces of a structure where the outdoor air is cleaned, temperature adjusted, and humidity adjusted by a Dedicated Outdoor Air System (DOAS) before entering the habitable spaces of a structure. The positively pressured indoor air is exhausted into the building envelope and subsequently to the exterior.

(191) Part A—The Mechanical System

(192) The volume of outside air entering the structure is as specified by ASHRAE and is based upon the building occupancy and use.

(193) Outside air is cleaned, filtered, and temperature and humidity adjusted through the DOAS for heating and cooling. This low volume/high velocity air is continuously injected into the building to regulate building temperature and humidity. Indoor air contamination is reduced since indoor air is not recirculated; rather exhausted out of the occupied spaces as recommended by ASHRAE. State of the art mechanical and ducting systems are currently available to economically achieve these goals. Further discussion of these mechanical and ducting systems are beyond the scope of this paper.

(194) Part B—The Building Envelope Ventilation

(195) SV indoor air is exhausted out of the occupied spaces and into the building envelope to regulate the temperature and humidity of the building envelope.

(196) Referring to FIG. **16A** a typical example of SV is shown. The HVAC system draws outside air into the unit where it is cleaned and temperature and humidity adjusted. This air is circulated within the building, exhausted into the attic, exhausted into the exterior walls, and released into the exterior environment. The attic and cavity walls are conditioned and ventilated with indoor air.

(197) The building, generally **1600** of FIG. **16A** is shown to include a roof **1602**, a 2" spray foam insulation **1604**, a solid 1× blocking **1606**, a 4" zero deflecto backflow preventer **1608**, and weep holes **1610**. The building **1600** is situated on a ground surface **1612** and supported by a foundation **1614**. The building **1600** also include an attic **1616**, a heating, cooling, dehumidification (HCD) system **1618**, and a wall air pathway **1620**. Unconditioned air **1630** enters through the HCD system **1618**. The HCD system **1618** cools or heats and dehumidifies unconditioned air **1630** and circulates conditioned air **1632** into an interior **1622** of the building **1600**. A portion of the conditioned air **1634** into the attic **1616**, flows through the preventer **1608**, flows through the wall air pathway **1620** as conditioned wall air **1636**, and exits out of the weep holes **1610** allowing moisture reduction in the walls and improved heating and cooling via the conditioned air **1636** flowing through the wall pathways **1620** and out the weep holes **1610**.

(198) Outside air is conditioned prior to distribution through the structure. Infiltration is eliminated. The building envelope is conditioned to near interior conditions. Differences for building retrofit or new construction include the absence of ventilation openings to the exterior (soffit vents, ridge vents, turtle vents) an updated HCc system and ducting (high velocity/low volume), the use of insulation beneath the roofing, the installation of a backflow preventer at top of wall, and the use of additional sealants at material terminations of the building envelope.

(199) Referring now to FIG. **16B**, a flooded structure equipt with SV is shown. The sheathing behind the brick veneer, when wet, can become contaminated and soft. The sheathing is a part of the shear walls to keep the building upright. In order to remove and replace the moisture damaged sheathing and conform to existing construction, removal and replacement of the brick veneer is necessary. This SV repair method allows the brick veneer to remain intact, provides drying of the structure for this and future events, and minimizes repair costs.

(200) The building, generally **1650**, of FIG. **16B** is shown to include a roof **1652**, a 2" spray foam insulation **1654**, a solid 1× blocking **1656**, a 4" zero deflecto backflow preventer **1658**, brick ties **1660**, mortar fill holes and cracks in the brick veneer **1662**, and weep holes **1664**. The building **1650** is situated on a ground surface **1666** and supported by a foundation **1668**. The building **1650** also include an attic **1670**, a HCD system **1672**, and a wall air pathway **1674**. The building **1650** also includes attic insulation **1676**, brick scaler **1678**, wall layers **1680**, a foil faced insulation layer **1682**, 2× solid blocking **1684**, and wall insulation **1686**. The HCD system **1672** cools or heats and dehumidifies unconditioned air **1690** and circulates conditioned air **1692** into an interior **1688** of the building **1650**. A portion of the conditioned air **1694** into the attic **1670**, flows through the preventer **1658**, flows through the wall air pathway **1620** as conditioned wall air **1696**, and exits out of the weep holes **1664** allowing moisture reduction in the walls and improved heating and cooling via the conditioned air **1696** flowing through the wall air pathways **1674** and out the weep holes **1664**.

(201) Moisture Limitations

(202) Many structural components are compromised by moisture. For example, mold can occur when the surface relative humidity of a building component exceeds 70%(±) and temperatures are above 4° C. (40° F.). Wood rot occurs at moisture contents of 30% or greater and with temperatures ranging between 10° C. (50° F.) and 40° C. (100° F.). Nail corrosion occurs at relative humidities greater than 60%; a maximum 20% wood moisture content is preferred to minimize nail corrosion within the wood. The Structural Board Association states that Exposure 1 OSB materials are

intended to be used in covered and well ventilated areas; this exposure relates to an OSB moisture content of 8% to 9% or a solid wood moisture content of 12% to 15%. Relative humidities above 80% are not recommended. ASHRAE, ANSI, and EPA concur that the relative humidity of interior ambient air should be between 30% and 60% with temperatures between 20° ° C. (68° F.) and 27° ° C. (81° F.) at all times.

(203) Excess moisture compromises structural components.

(204) WUFI Analyses

(205) WUFI, developed with the assistance of Oak Ridge National Laboratory (ORNL), analyzes the hygrothermal properties of building envelopes. WUFI allows realistic calculations of the transient coupled heat and moisture transport in multi-layer building envelopes based upon exterior weather data and normal interior climates. WUFI is an acronym for Wärme Und Feuchte Instationär, which translated means heat and moisture transiency.

(206) Referring now to FIG. 17, an office building was analyzed using WUFI. The occupant requires 1.6 Air Changes per Hour (ACH) located in Houston, Texas having the indicated dimensions. The office building, generally 1700, includes walls 1710. The walls 1710 includes a brick veneer 1712, an air gap 1714, a Tyvek® (moisture barrier) layer 1716, an OSB sheathing 1718 supported by stud framing, batt insulation 1720, and gypsum board supported by stud framing 1722.

(207) The office building 1700 also include a roof 1730. The roof 1730 includes a composite shingle roofing 1732, an OSB sheathing supported rafters 1734, a spray-on insulation layer for structure ventilation systems 1736, attic air 1738, batt insulation 1740, and gypsum board ceiling supported joists 1742.

(208) Attic Assembly

(209) Referring now to FIGS. 18A&B, plots of temperature and water content versus cross-section of attic air and ceiling insulation of a conventional HVAC system and attic air of a structural ventilation (SV) HVAC system of the office building of FIG. 17, respectively, are shown. The roof 1730 is at the top, and the ceiling 1742 is at the bottom.

(210) Looking at FIG. 18A, the plot of the conventional HVAC system temperature profile versus cross-section, generally, 1800, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally 1820.

(211) The plot 1800 displays a temperature line 1802 across the roof 1730, attic air 1738 in an attic interior 1739, the ceiling insulation 1740, and the gypsum board 1742.

(212) The plot 1820 displays a moisture content line 1822, an area representing the relative humidity of attic air 1824, an average attic air humidity line 1826, an area representing a moisture content of the attic air 1828, the interior attic air 1804, and the ceiling insulation 1806 across the roof 1730, attic air 1738 in an attic interior 1739, the ceiling insulation 1740, and the gypsum board 1742.

(213) Over a one year period, at the conventional system, the surface relative humidity exceeds 80%. The this humidity level mold growth would be expected, nail corrosion would be expected, and OSB moisture contents are elevated.

(214) Looking at FIG. 18B, the plot of a SV HVAC system temperature profile versus cross-section, generally, 1850, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally 1870.

(215) The plot 1850 displays an attic air temperature line 1852 measured across the roof 1730, the attic air 1738 in an attic interior 1739, the ceiling insulation 1740, and the gypsum board 1742.

(216) The plot 1870 displays a moisture content line 1872, an area 1874 representing the relative humidity of attic air 1738, an average attic air humidity line 1876, an area 1878 representing a moisture content of the attic air 1738 in an attic interior 1739, and the ceiling insulation 1740 measured across the roof 1730, attic air 1738 in an attic interior 1739, the ceiling insulation 1740, and the gypsum board 1742.

(217) At the SV system, these conditions do not exist except at the base of the roof insulation.

(218) Referring now to FIGS. **19A&B**, a plot of temperature values and relative humidity values over a year at the top surface of the ceiling insulation (0.18407 inch fiber glass insulation) in the conventional HVAC system and a plot of temperature values and relative humidity values over a year at the top surface of the ceiling insulation (0.204576 fiber glass insulation) in the SV HVAC system of the office building of FIG. **17**, respectively, are shown. With the SV system, relative humidity is reduced 35% in the summer months and the temperatures increase 15° C. (25° F.) in the winter months and decrease 12° C. (10° F.) in the summer months. The RSI-6 (R-35) assembly becomes RSI.sub.HT-38 (R.sub.HT-215). R.sub.HT is the Hygrothermal Resistance as defined herein.

(219) Wall Assembly

(220) Referring now to FIGS. **20A&B**, plots of temperature and water content versus cross-section of the wall assembly of a conventional HVAC system and a structural ventilation (SV) HVAC system of the office building of FIG. **17**, respectively, are shown measured across the brick veneer **1712** is on the left, the gypsum board **1724** is on the right, and the moisture barrier **1716**, the air gap **1714**, air **1715** in the air gap, the sheathing **1718**, and the insulation **1722** are interposed therebetween.

(221) Looking at FIG. **20A**, a plot of a conventional HVAC system temperature profile versus cross-section, generally, **2000**, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally **2020**.

(222) The plot **2000** displays a temperature line **2002** measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the air **1715** in the air gap, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**.

(223) The plot **2020** displays a wall assembly moisture content line **2022**, an area **2024** representing a moisture content measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**, an average wall assembly relative humidity line **2026** measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**, and an area **2028** representing a relative humidity measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**.

(224) Over a one year period, at the conventional system, exclusive of components outside the air cavity, the surface relative humidity exceeds 80%. Mold growth is expected. Nail corrosion is expected. OSB moisture contents are elevated.

(225) Looking at FIG. **20B**, a plot of a SV HVAC system temperature profile versus cross-section, generally, **2050**, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally **2070**.

(226) The plot **2050** displays a wall assembly temperature line **2052** measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**.

(227) The plot **2070** displays a wall assembly moisture content line **2072**, an area **2074** representing a moisture content measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**, an average relative humidity line **2076** measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**, and an area **2078** representing a relative humidity measured across the brick veneer **1712**, the moisture barrier **1716**, the air gap **1714**, the gap air **1715**, the sheathing **1718**, the insulation **1722**, and the gypsum board **1724**.

(228) At the SV system, these conditions do not exist.

(229) Referring now to FIGS. **21A&B**, plots of temperature values and relative humidity values

over a year at the exterior surface of the oriented strand board (OSB) sheathing in the conventional HVAC system (0.007039 inches) and the SV HVAC system (0.007039 inches) of the office building of FIG. 17, respectively, are shown, the temperatures and relative humidities of the conventional system and SV system, respectively, are shown. With the SV system, relative humidity is reduced 15% in the summer months; the temperatures increase 8° C. (15° F.) in the winter months and decrease 12° C. (10° F.) in the summer months. The RSI-3.5 (R-20) assembly becomes RSI.sub.HT-7 (R.sub.HT-40).

(230) Window Assembly

(231) Referring now to FIGS. 22A&B, plots of temperature and water content in the conventional system and the SV system versus cross-section of the window assembly of the office building of FIG. 17, respectively, are shown measured across the air gap 1714 from the left window to the right window.

(232) Looking at FIG. 22A, a plot of a conventional HVAC system temperature profile versus cross-section, generally, 2200, and a plot of a conventional HVAC system moisture content and humidity versus cross-section, generally 2220.

(233) The plot 2200 displays a temperature line 2202 measured across the air gap 1714.

(234) The plot 2220 displays a window assembly moisture content line 2222, an area 2224 representing a moisture content measured across the air gap 1714, an average window assembly relative humidity line 2226 measured across the air gap 1714, and an area 2228 representing a relative humidity measured across the air gap 1714.

(235) Looking at FIG. 22B, a plot of a SV HVAC system temperature profile versus cross-section, generally, 2250, and a plot of a SV HVAC system moisture content and humidity versus cross-section, generally 2270.

(236) The plot 2250 displays a wall assembly temperature line 2252 measured across the air gap 1714.

(237) The plot 2270 displays a wall assembly moisture content line 2272, an area 2274 representing a moisture content measured across the air gap 1714, an average relative humidity line 2276 measured across the air gap 1714, and an area 2278 representing a relative humidity measured across the air gap 1714.

(238) The interior glass surface humidity is lowered 50%. In this instance, it would be preferable to use insulated glass at the exterior pane to minimize condensation.

(239) Referring now to FIGS. 23A&B, the temperatures and relative humidities at the interior surface of the window of the conventional HVAC system and SV HVAC system, respectively, are shown. With the SV system, relative humidity is reduced 50% in the winter months; the temperatures increase 15° ° C. (25° F.) in the winter months and decreases 8° C. (15° F.) in the summer months. The RSI-0.2 (R-1 assembly becomes RSI.sub.HT-0.9 (R.sub.HT-5).

(240) If insulated glass was used at both panes, the thermal resistances would be additive. So, if each insulated pane is RSI-0.35 (R-2), $\Sigma R = \text{RSI.sub.HT-1.6}$ (R.sub.HT-9) of the window assembly. SV controls the building envelope temperatures and humidities.

(241) Additional WUFI Analyses

(242) The School Building, defined in FIG. 24, was also analyzed using WUFI. The building requires 3.1 ACH. Location: Houston, Texas ***

(243) Thermal Resistance (RSI and R) and Hygrothermal Resistance (RSI.sub.HT and R.sub.HT)

(244) The designation RSI (R) is the static thermal resistance of a component, or assembly, due to the physical characteristics of the materials.

(245) Hygrothermal Resistance, RSI.sub.HT (R.sub.HT) combines the static thermal resistance of the material with the dynamic coupled heat and moisture transport through the materials generated by the conditioned airflow through the building envelope.

(246) Hygrothermal Resistance increases with exhaust air cavity height/width and exhaust airflow

rate.

(247) Attics

(248) Referring now to FIGS. 25A&B, the R.sub.HT of the attic of the Office Building of FIG. 17 and the R.sub.HT of the attic of the School of FIG. 24, respectively, are shown. In each, static thermal resistance is approximately RSI-9 (R-50). With SV, the hygrothermal resistance increases significantly with greater attic height and greater air flow rates. Doubling the attic height increases R.sub.HT by a factor of 1.7. Doubling the air flow rate increases R.sub.HT by a factor of 1.7. At the Office Building, the Attic ACH=3 resulting in an RSI.sub.HT-35 (R.sub.HT-200); an increase by a factor of 4. At the School Building, the Attic ACH=4.7 resulting in an RSI.sub.HT-50 (R.sub.HT-275); an increase by a factor of 5.5.

(249) Walls

(250) Referring now to FIGS. 25C&D, the R.sub.HT of the walls of the Office Building FIG. 17 and the R.sub.HT of the walls of the School Building FIG. 24, respectively, are shown. In each, static thermal resistance is approximately RSI-3.5 (R-20). With SV, the hygrothermal resistance increases significantly with greater air flow rates. At the Office Building, southern exposure of the brick veneer wall results in little change. Doubling the air flow rate increases R.sub.HT by a factor of 1.6. The Cavity ACH=180 resulting in an RSI.sub.HT-7 (R.sub.HT-40); an increase by a factor of 2. At the School Building, southern exposure of the brick veneer wall results in an increase of R.sub.HT. Doubling the air flow rate increases R.sub.HT by a factor of 1.5. The Cavity ACH=585 resulting in an RSI.sub.HT-19 (R.sub.HT-110); an increase by a factor of 5.5.

(251) Windows

(252) Referring now to FIG. 26, the R.sub.HT of the windows of both buildings are shown. In each, static thermal resistance of a single pane clear window is approximately RSI-0.2 (R-1). Southern exposure and dark tint of the window results in a increase of R.sub.HT by a factor of 1.5.

(253) With SV, the hygrothermal resistance increases significantly with greater air flow rates. Doubling the air flow rate increases R.sub.HT by a factor of 1.3. At the Office Building, the Window ACH=180 resulting in RSI.sub.HT-0.9 (R.sub.HT-5). At the School Building, the Window ACH=585 resulting in RSI.sub.HT-1.4 (R.sub.HT-8).

(254) As stated earlier, R and R.sub.HT are additive. If we install thermal windows each side of the air stream that are each RSI-0.35 (R-2), the Office Building windows would attain RSI.sub.HT-1.6 (R.sub.HT-9). The School Building windows would attain RSIHT=2.1 (R.sub.HT-12). Each increase is significant.

(255) SV significantly increases the hygrothermal resistance of the building envelope.

(256) HVAC Sizing

(257) The Office Building was analyzed with Manual J based upon the lowest RHT of the winter months at various locations with-in the United States.

(258) Referring now to FIG. 27, the heating and cooling requirements of the building in various cities are shown. At each city, the conventional heating and cooling requirements are shown as the tall orange and blue bars. The SV heating and cooling requirements are shown as the short orange and blue bars. The grey bars are the percent SV unit sizing is of conventional unit sizing. The percentages range from 69% to 75%.

(259) SV results in smaller equipment sizes regardless of location.

(260) Energy Usage for Heating

(261) The Office Building was analyzed with the WUFI Post Processor based upon the RHT of winter monthly averages that are based upon hourly calculations. Typically, the RHT is greater before and after the coldest month. Referring now to FIG. 28, the annual heating energy usage of the building at several cities are shown. The conventional heating and SV heating requirements are presented as separate groups. The SV group shows percentages in the bars that indicates the percent SV energy usage is of conventional usage. The percentages range from 49% to 71%.

(262) SV results in significant energy savings regardless of location.

(263) The School Building was also analyzed with the WUFI Post Processor. Referring now to FIG. 29, the annual heating energy usage of the school building at several cities are shown. The conventional heating and SV heating requirements are presented as separate columns. The SV columns show percentages in the bars that indicates the percent SV energy usage is of conventional usage. The percentages range from 66% to 75%.

(264) SV results in significant energy savings regardless of location.

(265) Moisture Issues

(266) Moisture infiltration is a common problem with building envelopes. Moisture, typically from the exterior environment, passes through defective joints in the exterior finishes and roofing. The joints typically occur at material terminations such as window and door perimeters, roofing and wall penetrations, roof to wall interfaces, and the like.

(267) As noted earlier, moisture can also develop due to naturally occurring moisture due to hygrothermal activity within the building envelope. This is most prevalent when low permeability materials are installed.

(268) The following building envelope moisture content readings were compared to WUFI analyses. Red is well above expected values, yellow is slightly above expected values, green similar to expected values, and blue is below expected values.

(269) Expected moisture content readings of interior components were determined by air temperature and humidity readings.

(270) Referring now to FIGS. 30A-C, moisture content readings at the walls of a one-story, one-year old, daycare are shown. Gypsum sheathing was measured at the exterior faces of the wall studs and gypsum board was measured at the interior of the structure. The interior gypsum board finish moisture contents are elevated at the perimeter of the building. Window leaks are evident. The gypsum sheathing behind the brick veneer is very wet.

(271) In this instance, the weep holes at the base of the wall were plugged with mortar, rainwater was seeping behind the veneer, and the windows leaked.

(272) Referring now to FIGS. 31A-C, moisture content readings at the third floor walls of a three-story, three-year old, house are shown. Gypsum board, baseboards, and windowsills were measured at the interior of the structure.

(273) The gypsum board finish moisture contents are moderate at the perimeter walls and the interior walls. The baseboards are typically moist beneath windows and at doors. Windowsill reading are elevated.

(274) Window and door leakage are the prime contributors. Moisture is collecting within the building envelope.

(275) Referring now to FIGS. 32A-D, moisture content readings at the third floor walls and ceiling of a three-story, four-year old, house are shown. Gypsum board, baseboards, windowsills, and the ceiling were measured at the interior of the structure. The interior gypsum board moisture contents are elevated at the central area of the house and some exterior walls. The baseboards are more moist at the interior of the building as opposed to the perimeter. Windowsill readings are elevated. The ceiling is typically moist with greater moisture at the right side of the drawing.

(276) Moisture is collecting within the attic due to the lack of adequate ventilation. Moisture from this source is infiltrating the interior and perimeter walls of the upper floor. Window and door leakage are also issues.

Embodiments of the Disclosure

(277) Embodiment 1. A structure comprising: a floor; a roof; one or more exterior walls, each exterior wall including: an exterior wall air path way including: one or more exterior wall air pathway inlets, and one or more exterior wall air pathway outlets; a plurality of interior walls; one or more of exterior window assemblies, each of the one or more of exterior window assemblies including: an exterior window, an interior window, and a window air pathway interposed between the exterior window and the interior window and including: a window air flow pathway inlet, and a

window air flow pathway outlet, a plurality of interior spaces defined by the exterior and interior walls; an HVAC unit including: one or more HVAC inlet conduits, each of the one or more HVAC inlet conduits including: an HVAC inlet including: an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit; one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including: an HVAC exhaust inlet, and an HVAC exhaust outlet configured to support an air flow of the exhaust conditioned air out of the structure, to the HVAC inlet, or any combination thereof; one or more HVAC interior space outlet conduits, each of the one or more HVAC interior space outlet conduits including: an HVAC interior space outlet including: an HVAC interior space outlet control valve configured to control a supply of conditioned air to each of the interior spaces; one or more HVAC exterior wall conduits, each of the one or more HVAC exterior wall conduits including: an HVAC exterior wall outlet including: an HVAC exterior wall control valve configured to support an exterior wall pathway air flow of conditioned air to each of the exterior wall pathways, wherein wall pathway air flows reduce, remove, and/or eliminate moisture in the exterior walls and increase heating and cooling efficiencies, and an HVAC window conduit for each of the one or more of exterior window assemblies, each HVAC window conduit including: an HVAC window conduit inlet, and an HVAC window conduit inlet, wherein window pathway air flows reduce, remove, and/or eliminate moisture in the exterior windows and increase heating and cooling efficiencies.

(278) Embodiment 2. The structure of Embodiment 1, wherein: one, some, or all of the exterior wall air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets, and/or one, some, or all of the exterior window air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

(279) Embodiment 3. The structure of Embodiment 1, wherein each of the plurality of interior walls includes: one or more interior wall air flow pathways, each of the one or more interior wall air flow pathways including: a interior wall air pathway inlet, and a interior wall air pathway outlet; and the HVAC unit further including: one or more HVAC interior wall conduits, each of the one or more HVAC interior wall conduits including: an HVAC interior wall outlet including: an HVAC interior wall control valve configured to support an interior wall pathway air flow of conditioned air to each of the interior wall pathways, wherein the interior air flows reduce, remove, and/or eliminate moisture in the interior walls, and one, some, or all of the interior wall air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.

(280) Embodiment 4. The structure of Embodiment 1, wherein each of the one or more exterior walls includes: an outer surface including: an insulating layer; a weather-resistant layer disposed on an outer surface of the insulating layer, a sheathing, a liquid barrier, or any combination thereof, and/or an inner surface including: an insulating layer, a weather-resistant layer disposed on an outer surface of the insulating layer, a sheathing, a liquid barrier, or any combination thereof.

(281) Embodiment 5. The structure of Embodiment 1, further comprising: an air flow control system including: a processing unit; and plurality of sensors; wherein one, some, or all of the control valves are active control valves, wherein the processing unit is in communication with the active control valves, and wherein the processing unit is configured to: receive input from the sensors, and control the HVAC unit, and control the active control valves so that air flow rates are controlled into the interior spaces and into the pathways, and during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

(282) Embodiment 6. The structure of Embodiment 5, wherein the sensors comprise temperature

sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

(283) Embodiment 7. The structure of Embodiment 5, wherein one or more of the sensors are disposed in the air flow pathways, in the interior spaces, or in both.

(284) Embodiment 8. The structure of Embodiment 5, wherein: all of the control valves are active control valves, and during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

(285) Embodiment 9. The structure of Embodiment 8, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

(286) Embodiment 10. The structure of Embodiment 1, further comprising: a dispensing assembly associated with one, some, or all of the pathways and including: a reservoir containing a treating composition; a dispensing supply conduit having: a dispensing assembly control valve, a dispensing head at its distal end, wherein: the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes or mixtures thereof; the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

(287) Embodiment 11. A multistory structure comprising: a plurality of stories, each of the stories including: a floor, a ceiling, and a dead space disposed above the ceiling; one or more exterior walls, each of the one or more exterior walls including: an exterior wall air flow pathway; and at least one exterior wall exhaust conduit; a plurality of exterior window assemblies, each of the plurality of exterior window assemblies including: an exterior window, an interior window, and a window air pathway interposed between the exterior window and the interior window and including: a window air flow pathway inlet, and a window air flow pathway outlet, a plurality of interior walls, each of the interior walls including: an interior wall air flow pathway; and at least one interior wall outlet conduit; a plurality of interior spaces defined by the one or more exterior walls and interior walls; an air handling system comprising: an HVAC unit; an HVAC inlet conduit including: an HVAC surroundings inlet conduit having: a surrounding inlet including: a surrounding inlet active control valve configured to supply unconditioned air to the HVAC unit; a return conditioned air conduit including: a return conduit active control valve configured to control an amount of exhausted conditioned air to the HVAC unit; a plurality of floor return conduits, each of the floor return conduits including: an inlet disposed in the each of the floor dead spaces including: a floor return conduit active control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the dead spaces; a main HVAC outlet conduit including: a plurality of floor outlet conduits, each of the floor outlet conduits including: a floor outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the floors; a plurality of interior space outlet conduits, each of the interior space outlet conduits having: a plurality of interior space outlets, each of the interior space outlets

having an interior space outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the interior space outlets; and exterior window outlet conduit active control valves configured to control an amount of conditioned air supplied to each of the plurality of exterior windows; one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including: a HVAC exhaust conduit active control valve configured to control an amount of conditioned air exhausted to the surroundings; and an air flow control system including: a processing unit coupled to or associated with a plurality of sensors, the active control valves and the air handling system; one or more of the sensors is disposed in each of the interior spaces; one or more of the sensors is disposed in the pathways; wherein the processing unit is configured to: receive input from the sensors; and control the HVAC unit and the active control valves based on the sensor inputs to control unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and pathway air flow rates through the pathways and into the surroundings.

(288) Embodiment 12. The structure of Embodiment 11, wherein each of the floor return conduits further including: at least one return air active control valve configured to control an amount of the return conditioned air to the return conditioned air conduit.

(289) Embodiment 13. The structure of Embodiment 11, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

(290) Embodiment 14. The structure of Embodiment 13, wherein, during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway based the received sensor input so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

(291) Embodiment 15. The structure of Embodiment 14, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

(292) Embodiment 16. The structure of Embodiment 11, further comprising: a dispensing assembly associated with one, some, or all of the pathways and including: a reservoir containing a treating composition; a dispensing supply conduit including: a dispensing assembly control valve, a dispensing head at its distal end, wherein: the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes, or mixtures thereof; the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

CLOSING PARAGRAPH OF THE DISCLOSURE

(293) All references cited herein are incorporated by reference. Although the disclosure has been disclosed with reference to its preferred embodiments, from reading this description those of skill in the art may appreciate changes and modification that may be made which do not depart from the scope and spirit of the disclosure as described above and claimed hereafter.

Claims

1. A structure comprising: a floor; a roof; one or more exterior walls, each exterior wall including: an exterior wall air pathway including: one or more exterior wall air pathway inlets, and one or more exterior wall air pathway outlets; a plurality of interior walls; one or more of exterior window assemblies, each of the one or more of exterior window assemblies including: an exterior window, an interior window, and a window air pathway interposed between the exterior window and the interior window and including: a window air flow pathway inlet, and a window air flow pathway outlet, a plurality of interior spaces defined by the exterior and interior walls; an heating, ventilation, and air conditioning (HVAC) unit including: one or more HVAC inlet conduits, each of the one or more HVAC inlet conduits including: an HVAC inlet including: an HVAC inlet control valve configured to control a supply of unconditioned air or a mixture of unconditioned air and exhaust conditioned air to the HVAC unit; one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including: an HVAC exhaust inlet, and an HVAC exhaust outlet configured to support an air flow of the exhaust conditioned air out of the structure, to the HVAC inlet, or any combination thereof; one or more HVAC interior space outlet conduits, each of the one or more HVAC interior space outlet conduits including: an HVAC interior space outlet including: an HVAC interior space outlet control valve configured to control a supply of conditioned air to each of the interior spaces; one or more HVAC exterior wall conduits, each of the one or more HVAC exterior wall conduits including: an HVAC exterior wall outlet including: an HVAC exterior wall control valve configured to support an exterior wall pathway air flow of conditioned air to each of the exterior wall pathways, wherein wall pathway air flows reduce, remove, and/or eliminate moisture in the exterior walls and increase heating and cooling efficiencies, and an HVAC window conduit for each of the one or more of exterior window assemblies, each HVAC window conduit including: an HVAC window conduit inlet, and an HVAC window conduit inlet, wherein window pathway air flows reduce, remove, and/or eliminate moisture in the exterior windows and increase heating and cooling efficiencies.
2. The structure of claim 1, wherein: one, some, or all of the exterior wall air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets, and/or one, some, or all of the exterior window air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.
3. The structure of claim 1, wherein each of the plurality of interior walls includes: one or more interior wall air flow pathways, each of the one or more interior wall air flow pathways including: an interior wall air pathway inlet, and an interior wall air pathway outlet; and the HVAC unit further including: one or more HVAC interior wall conduits, each of the one or more HVAC interior wall conduits including: an HVAC interior wall outlet including: an HVAC interior wall control valve configured to support an interior wall pathway air flow of conditioned air to each of the interior wall pathways, wherein the interior air flows reduce, remove, and/or eliminate moisture in the interior walls, and one, some, or all of the interior wall air flow pathway outlets include: a passive one-way or unidirectional valve or an active control valve to stop air or water flow into each of the exterior wall air flow pathway outlets.
4. The structure of claim 1, wherein each of the one or more exterior walls includes: an outer surface including: an insulating layer; a weather-resistant layer disposed on an outer surface of the insulating layer, a sheathing, a liquid barrier, or any combination thereof, and/or an inner surface including: an insulating layer, a weather-resistant layer disposed on an outer surface of the insulating layer, a sheathing, a liquid barrier, or any combination thereof.
5. The structure of claim 1, further comprising: an air flow control system including: a processing unit; and plurality of sensors; wherein one, some, or all of the control valves are active control

valves, wherein the processing unit is in communication with the active control valves, and wherein the processing unit is configured to: receive input from the sensors, and control the HVAC unit, and control the active control valves so that air flow rates are controlled into the interior spaces and into the pathways, and during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

6. The structure of claim 5, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

7. The structure of claim 5, wherein one or more of the sensors are disposed in the air flow pathways, in the interior spaces, or in both.

8. The structure of claim 5, wherein: all of the control valves are active control valves, and during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

9. The structure of claim 8, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

10. The structure of claim 1, further comprising: a dispensing assembly associated with one, some, or all of the pathways and including: a reservoir containing a treating composition; a dispensing supply conduit having: a dispensing assembly control valve, a dispensing head at its distal end, wherein: the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes or mixtures thereof; the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.

11. A multistory structure comprising: a plurality of stories, each of the stories including: a floor, a ceiling, and a dead space disposed above the ceiling; one or more exterior walls, each of the one or more exterior walls including: an exterior wall air flow pathway; and at least one exterior wall exhaust conduit; a plurality of exterior window assemblies, each of the plurality of exterior window assemblies including: an exterior window, an interior window, and a window air pathway interposed between the exterior window and the interior window and including: a window air flow pathway inlet, and a window air flow pathway outlet, a plurality of interior walls, each of the interior walls including: an interior wall air flow pathway; and at least one interior wall outlet conduit; a plurality of interior spaces defined by the one or more exterior walls and interior walls; an air handling system comprising: an heating, ventilation, and air conditioning (HVAC) unit; an HVAC inlet conduit including: an HVAC surroundings inlet conduit having: a surrounding inlet including: a surrounding inlet active control valve configured to supply unconditioned air to the HVAC unit; a return conditioned air conduit including: a return conduit active control valve configured to control an amount of exhausted conditioned air to the HVAC unit; a plurality of floor return conduits, each of the floor return conduits including: an inlet disposed in the each of the floor dead spaces including: a floor return conduit active control valve configured to control an amount of exhaust conditioned air supplied to the return conduit from each of the dead spaces; a

main HVAC outlet conduit including; a plurality of floor outlet conduits, each of the floor outlet conduits including: a floor outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the floors; a plurality of interior space outlet conduits, each of the interior space outlet conduits having: a plurality of interior space outlets, each of the interior space outlets having an interior space outlet conduit active control valve configured to control an amount of conditioned air supplied to each of the interior space outlets; and exterior window outlet conduit active control valves configured to control an amount of conditioned air supplied to each of the plurality of exterior windows; one or more HVAC exhaust conduits, each of the one or more HVAC exhaust conduits including: a HVAC exhaust conduit active control valve configured to control an amount of conditioned air exhausted to the surroundings; and an air flow control system including: a processing unit coupled to or associated with a plurality of sensors, the active control valves and the air handling system; one or more of the sensors is disposed in each of the interior spaces; one or more of the sensors is disposed in the pathways; wherein the processing unit is configured to: receive input from the sensors; and control the HVAC unit and the active control valves based on the sensor inputs to control unconditioned air flow rates into the HVAC, interior space air flow rates from the HVAC, return air flow rates into the HVAC, exhaust air flow rate to the surroundings, and pathway air flow rates through the pathways and into the surroundings.

12. The structure of claim 11, wherein each of the floor return conduits further including: at least one return air active control valve configured to control an amount of the return conditioned air to the return conditioned air conduit.

13. The structure of claim 11, wherein the sensors comprise temperature sensors, water/moisture sensors, air flow sensors, chemical sensors, other physical sensors, and combinations thereof.

14. The structure of claim 13, wherein, during a 24 hour period of time, the processing unit is further configured to: adjust the air flow into each of the interior spaces and into each pathway based the received sensor input so that one or more interior space properties and/or one or more pathway properties are maintained at desired values.

15. The structure of claim 14, wherein the one or more properties include moisture concentrations in the interior spaces, moisture concentrations in the pathways, temperatures in the interior spaces, temperatures in the pathways, pressures in the interior spaces, pressures in the pathways, air flow rates in the interior spaces, air flow rates in the pathways, sensed chemical concentrations in the interior spaces, sensed chemical concentrations in the pathways, microbial concentrations in the interior spaces, microbial concentrations in the pathways or mixtures and combinations thereof.

16. The structure of claim 11, further comprising: a dispensing assembly associated with one, some, or all of the pathways and including: a reservoir containing a treating composition; a dispensing supply conduit including: a dispensing assembly control valve, a dispensing head at its distal end, wherein: the dispensing assembly control valve is configured to control an amount of treating solution or gas introduced into one, some, or all of the pathways, the treating composition comprising a mold control composition, a pesticide, an insecticide, a fungicide, a bactericide, a rodent poison, other control compositions, a gas, a fire retardant compositions, a foam, or mixtures or combinations thereof; the gas comprising nitrogen, nitrogen enriched air, argon, haloalkanes, or mixtures thereof; the haloalkanes comprising iodomethane; bromomethane, bromochloromethane, carbon tetrachloride, tribromofluoromethane, dichlorofluoromethane, bromodifluoromethane, dibromodifluoromethane, bromochlorodifluoromethane, dichlorodifluoromethane, bromotrifluoromethane, tetrafluoromethane, 1,2-dichlorotetrafluoroethane, dibromotetrafluoroethane, hexafluoroethane, or mixtures thereof.
